- The USA is the world's largest producer of nuclear power, accounting for about 30% of worldwide generation of nuclear electricity.
- The country's nuclear reactors produced 816 TWh in 2024, 18% of total electrical output.
- Vogtle 3 was connected to the grid in April 2023, followed by unit 4 in March 2024.
- The Inflation Reduction Act was signed into law in August 2022. The Act provides support for existing and

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new nuclear development through investment and tax incentives for both large existing nuclear plants and newer advanced reactors, as well as high-assay low enriched uranium (HALEU) and hydrogen production.

- Targets to quadruple US nuclear capacity (to 400 GWe) by 2050 and to deploy advanced reactors in the near term were set by the administration in 2025.

94	0	41
OPERABLE REACTORS	REACTORS UNDER CONSTRUCTION	REACTORS SHUTDOWN
96,952 MWe	0 MWe	20,017 MWe

Electricity sector

Total generation (in 2023): 4453 TWh

Generation mix: natural gas 1865 TWh (42%); nuclear 808 TWh (18%); coal 742 TWh (17%); wind 426 TWh (10%); hydro 269 TWh (6%); solar 214 TWh (5%); biofuels & waste 61.4 TWh; oil 33.1 TWh; geothermal 19.5 TWh.

Import/export balance: 18.9 TWh net import (38.9 TWh imports; 20.0 TWh exports)

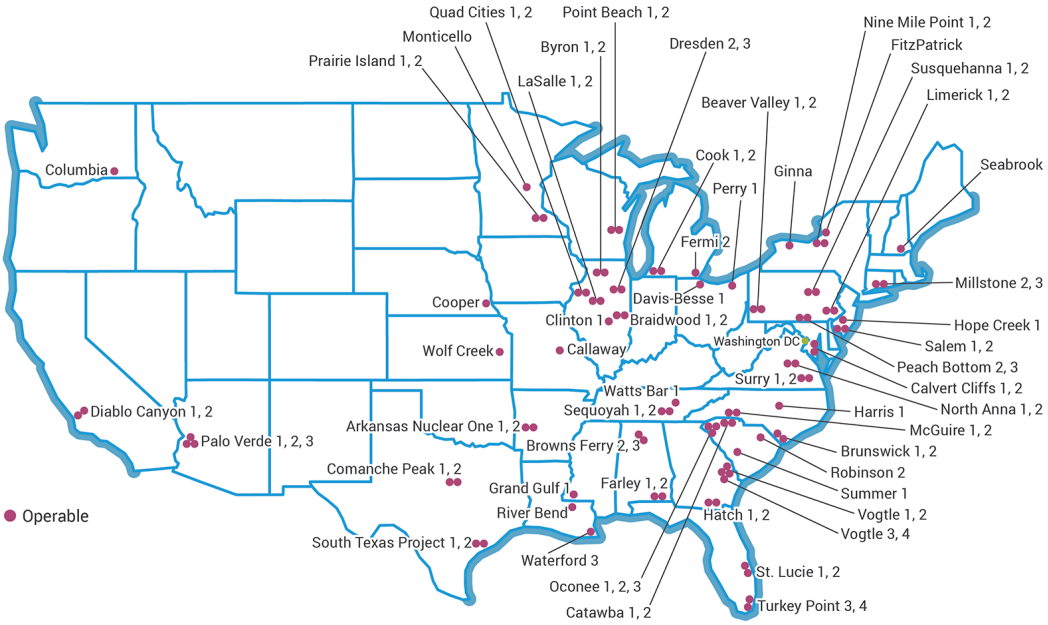
Total consumption: 3894 TWh

Per capita consumption: c. 11,600 kWh

Source: International Energy Agency and The World Bank. Data for year 2023.

Nuclear power industry

Operable reactors in the USA



A table of operable plants in the USA is available as an appendix to this page.

Nuclear power plays a major role in electricity provision across the country. The US fleet is operated by 30 different power companies across 30 different states. Since 2001 these plants have achieved an average capacity factor of over 90%. The average capacity factor has risen from 50% in the early 1970s, to 70% in 1991, and it passed 90% in 2002, remaining at around this level since. In 2019 it was a record 94%. The industry invests about $8 billion per year in maintenance and upgrades of the plants.

Nuclear plants generate nearly 20% of the nation's electricity overall and about 55% of its carbon-free electricity.

Since about 2010 sustained low natural gas prices dampened plans for new nuclear capacity (see section on New nuclear capacity below). However, in August 2022 the Inflation Reduction Act was passed by the US House of Representatives and later that month signed into law by President Joe Biden. The energy provisions in the Act outline support for existing and new nuclear development through investment and tax incentives for both large existing nuclear plants and newer advanced reactors, as well as HALEU and hydrogen production.

Later, in May 2025 US President Donald Trump signed a series of executive orders designed to stimulate growth in the industry with the aim of quadrupling capacity by 2050 (see section on Presidential executive orders, May 2025 below).

Nuclear industry development

The USA was a pioneer of nuclear power development.[a] Westinghouse designed the first fully commercial pressurized water reactor (PWR), a unit of 250 MWe capacity, Yankee Rowe, which started up in 1960 and operated to 1992. Meanwhile the boiling water reactor (BWR) was developed by the Argonne National Laboratory, and the first commercial plant, Dresden 1 (250 MWe) designed by General Electric, was started up in 1960. A prototype BWR, Vallecitos, ran from 1957 to 1963.

By the end of the 1960s, orders were being placed for PWR and BWR reactor units of more than 1000 MWe capacity, and a major construction program got under way. These remain practically the only types built commercially in the USA.[b]

Almost all the US nuclear generating capacity comes from reactors built between 1967 and 1990. Until 2013 there had been no new construction starts since 1977, largely because for a number of years gas generation was considered more economically attractive and because construction during the 1970s and 1980s had frequently been delayed.

Nuclear developments in USA suffered a major setback after the 1979 Three Mile Island accident, though that actually validated the very conservative design principles of Western reactors, and no-one was injured or exposed to harmful radiation. Many orders and projects were cancelled or suspended, and the nuclear construction industry went into the doldrums for two decades. Nevertheless, by 1990 over 100 commercial power reactors had been commissioned.

A further PWR – Watts Bar 2 – started up in 2016 following Tennessee Valley Authority's (TVA's) decision in 2007 to complete the construction of the unit, and Vogtle 3&4 were connected to the grid in 2023 and 2024 respectively.

Most reactors were built by regulated utilities, often state-based, which meant that they put the capital cost (whatever it turned out to be after, for example, delays) into their rate base and amortized it against power sales. Their consumers bore the risk and paid the capital cost. (With electricity deregulation in some states, the shareholders bear any risk of capital overruns and power is sold into competitive markets.)

Despite a near halt in new construction for more than 30 years, US reliance on nuclear power has grown. In 1980, nuclear plants produced 251 TWh, accounting for 11% of the country's electricity generation. In 2023 output was 779 TWh providing about 19% of electricity. Much of the increase came from the 47 reactors, all

approved for construction before 1977, that came online in the late 1970s and 1980s, more than doubling US nuclear generation capacity. The US nuclear industry has also achieved remarkable gains in power plant utilization through improved refuelling, maintenance and safety systems at existing plants. Average nuclear generation costs have come down from $52.83/MWh (in 2023 dollars) in 2012 to $31.76/MWh in 2023. This 40% reduction in nuclear generating costs since 2012 has been driven by: a 41% decrease in fuel costs; a 51% decrease in capital expenditures; and a 33% decrease in operating costs.[9]

Operationally, from the 1970s the US nuclear industry dramatically improved its safety and operational performance, and by the turn of the century it was among world leaders, with average net capacity factor over 90%.

This performance was achieved as the US industry continued deregulation, begun with passage of the Energy Policy Act in 1992. Changes accelerated after 1998, including mergers and acquisitions affecting the ownership and management of nuclear power plants.

Westinghouse bankruptcy

Westinghouse filed for Chapter 11 bankruptcy reorganization on 29 March 2017, after struggling to find cash to fund growing cost overruns at its two US nuclear plant projects (see above). The company listed assets of $4.3 billion and liabilities of $9.4 billion in the filing, and asked permission to pay about $50 million in employee salaries and benefits as well as $87.3 million to critical vendors during bankruptcy proceedings. Westinghouse and 30 affiliated companies filed for bankruptcy protection, listing about 35,000 creditors involved. Westinghouse said that its operations in Asia, Europe, the Middle East and Africa were not affected by the bankruptcy filings. Interim financing of $800 million was provided by Westinghouse parent company Toshiba and a New York private equity company, Apollo Capital Management. Toshiba said that it anticipated a new entity to be found by Westinghouse would take a leading role in bringing that company out of bankruptcy, and that its own control of Westinghouse had ended.

Westinghouse said its largest creditors were US construction company Fluor Enterprises – which was brought into the US nuclear plant projects in 2015 to take over construction management, and Chicago Bridge & Iron – in connection with the acquisition by Westinghouse of CB&I's Stone & Webster construction business in

late 2014. Fluor was owed almost $194 million, and CB&I $145 million. In March Toshiba said it would not provide additional funding without collateral, according to the bankruptcy protection filing. That resulted in the development of the debtor-in-possession financing, under which Westinghouse funded continuing operations. Westinghouse said it would work with the several owners of the nuclear plant projects in Georgia and South Carolina to “explore the continued feasibility of those projects in a manner that is cost-neutral and cash-neutral" to Westinghouse and its affiliates. Those owners of the Vogtle and Summer plants agreed to pay costs to continue construction themselves for a transition and evaluation period while final arrangements on future plant work were developed. The project at Summer has since been abandoned.

Westinghouse said that it remained committed to the AP1000 technology and would continue to support plants that were then being built in China, and planned for China, USA, India, Turkey, the UK and elsewhere. Its nuclear fuel business had revenues of $1.48 billion in fiscal 2015 (to end March 2016), and its operating plant business had revenues of $1.65 billion in the same period, while the new nuclear plant services business lost money.

Reactor lifetime extensions and regulation

The Nuclear Regulatory Commission (NRC) is the government agency established in 1974 to be responsible for regulation of the nuclear industry, notably reactors, fuel cycle facilities, materials and waste (as well as other civil uses of nuclear material).

In an historic move, the NRC in March 2000 renewed the operating licences of the two-unit Calvert Cliffs nuclear power plant for an additional 20 years. In March 2019 the NRC renewed the licence for Seabrook, extending the unit’s operation by 20 years to 2050. This took the number of US power reactors that have renewed their licences to 94, several of which have since shut down. Hence, almost all of the US power reactors are likely to be licensed to operate for 60 years, with owners undertaking major capital works to upgrade them at around 30-40 years. The licence renewal process typically costs $16-25 million, and the procedures for such renewals, with public meetings and thorough safety review, are exhaustive.

The original 40-year licences were always intended to be renewed in 20-year increments, as the 40-year period was more to do with amortization of capital rather than implying that reactors were

designed for only that operational lifespan. It was also a conservative measure, and experience since has identified life-limiting factors and addressed them. The NRC is now considering applications for the extension of operating licences beyond 60 out to 80 years, with its subsequent licence renewal (SLR) programme. As of January 2026:

- Reactors approved (to 80 years): Turkey Point 3&4, Peach Bottom 2&3, Surry 1&2, North Anna 1&2, Monticello, Oconee 1-3, Virgil C. Summer 1, Point Beach 1&2, Browns Ferry 1-3, Dresden 2&3.
- Reactors under review: St. Lucie 1&2, H.B. Robinson 2, Edwin I. Hatch 1&2.

The licence extensions to 60 years and beyond meant that major mid-life refurbishment, such as replacement of steam generators and upgrades of instrument and control systems, could be justified. While active plant components such as pumps and valves are under continuous scrutiny for operability, passive components need to be assessed for ageing which may have weakened them. There are R&D programmes focusing on this run by the Department of Energy (DOE), Electric Power Research Institute (EPRI), and American Society of Mechanical Engineers (ASME).

The NRC's reactor oversight and assessment process yields publicly-accessible information on the performance of plants in 19 key areas (14 indicators on plant safety, two on radiation safety and three on security). Performance against each indicator is reported quarterly on the NRC website according to whether it is normal, attracting regulatory oversight, provoking regulatory action, or unacceptable (in which case the plant would probably be shut down).

On the industry side, the Institute of Nuclear Power Operations (INPO) was formed after the Three Mile Island accident in 1979, to establish standards of performance against which individual plants could be regularly measured. An inspection of each member plant is typically performed every 18 to 24 months.

Following the accident at Japan's Fukushima nuclear plant in March 2011, which was exacerbated by inadequate outside assistance to the flooded reactors, the US nuclear industry set up the 'FLEX' accident response strategy. It has 61 centres across the country and two national centres which together provide the capacity to respond to nuclear power plant accidents anywhere in the country within 24 hours.

In September 2025 the USA and UK signed the Technology Prosperity Deal that deepens collaboration between the NRC and the UK's Office for Nuclear Regulation (ONR) and Environment Agency to support the completion of reactor design reviews within two years and site approvals within one year.

New nuclear capacity

From 1992 to 2005, some 270,000 MWe of new gas-fired plant was built, and only 14,000 MWe of new nuclear and coal-fired capacity came online. But coal and nuclear supplied almost 70% of US electricity at the time and provided price stability. When investment in these two technologies almost disappeared, unsustainable demands were placed on gas supplies and prices quadrupled, forcing large industrial users of it offshore and pushing gas-fired electricity costs towards 10 ¢/kWh. However, due to technological advances (in particular, horizontal drilling techniques) in shale gas fracking in the 2000s, costs became much lower.

The reason for investment being predominantly in gas-fired plant was that it offered the lowest investment risk. Several uncertainties inhibited investment in capital-intensive new coal and nuclear technologies. The average age of US nuclear plants is over 40 years, and major investment is also required in transmission infrastructure. This creates an energy investment crisis which is recognized in Washington, along with an increasing bipartisan consensus on the strategic importance and clean air benefits of nuclear power in the energy mix.

In 2002, the Department of Energy (DOE) launched its Nuclear Power 2010 (NP2010) programme to reinvigorate the US nuclear power industry through identifying sites, demonstrating untested regulatory processes and developing advanced nuclear technologies.

As part of the NP2010 initiative, in 2003, the Department of Energy (DOE) called for combined construction and operating licence (COL) proposals on the basis that it would fund up to half the cost of any accepted.

Several industry consortia were created for the purpose of preparing COL applications for new reactors. Although few of these projects went ahead, the secured COLs have a validity spanning 40 years from the date of issue, with the possibility of extension for a further 20 years.

Further information is given in Nuclear Power in the USA Appendix 3: COL Applications.

The Energy Policy Act (EPA) of 2005 provided a much-needed stimulus for investment in electricity infrastructure including nuclear power.

The EPA introduced a production tax credit (PTC) of 1.8 cents per kilowatt hour of electricity produced by new nuclear plants in their first eight years of operation. The tax credit was available only for the first 6000 MWe of new nuclear capacity and could not be claimed until electricity generation commenced.

Under the terms of the EPA, to qualify for the nuclear PTC, a plant needed to be in service on or before 31 December 2020, and the maximum value of the nuclear PTC was $6 billion over eight years (or $750 million per year). However in February 2018, an extension to the PTC was passed by the US Senate and Congress that allowed reactors entering service after 31 December 2020 to qualify for the tax credits, and allowed the US Energy Secretary to allocate credit for up to 6000 MWe of new nuclear capacity which enters service after 1 January 2021.

The Inflation Reduction Act of 2022 further extended the PTC, offering a credit of 1.5 cents per kilowatt hour for certain existing nuclear plants and plants placed in service after 31 December 2024. It runs through 2032. The credit, which is indexed to inflation and dependent on certain wage requirements being met, gradually declines as power prices rise above 2.5 cents per kilowatt hour.

For further discussion see information page on US Nuclear Power Policy.

New reactor construction got under way from 2012, with two units at the Vogtle nuclear power plant, and two units at the Summer nuclear power plant.*

* The project at Summer was subsequently cancelled.

Continued low gas prices have since depressed the prospects for commitment to further construction, and it is generally considered that natural gas prices need to recover to $8/GJ or /MMBtu before there is renewed confidence in deregulated states. In regulated states, a longer-term outlook is possible. Small modular reactors provide possible relief from major upfront finance burdens, but many of these designs are some way off having design certification from the NRC.

There are three regulatory initiatives that have enhanced the prospects of building new plants. First is the design certification process, second is provision for early site permits (ESPs) and third is the combined construction and operating licence (COL) process ('Part 52' of Title 10 of the Code of Federal Regulations) as an alternative to the 'Part 50' two-step process of construction permit followed by operating licence. All have some costs shared by the DOE.

A COL application under 10 CFR Part 52 may reference a design certification and/or an ESP. This allows the issues resolved during the design certification and ESP processes to be precluded from reconsideration during the COL licensing process.

Government support

Presidential executive orders, May 2025

In May 2025 US President Donald Trump signed a series of executive orders titled Reinvigorating the Nuclear Industrial Base, Reforming Nuclear Reactor Testing at the Department of Energy and Ordering the Reform of the Nuclear Regulatory Commission with the goal of "re-establishing the United States as the global leader in nuclear energy." The aim of the executive orders is to increase capacity from 100 GWe to 400 GWe by 2050, including the Department of Energy (DOE) prioritizing work "with the nuclear energy industry to facilitate 5 gigawatts of power uprates to existing nuclear reactors and have 10 new large reactors with complete designs under construction by 2030."

The executive order Reinvigorating the Nuclear Industrial Base states:

> "Swift and decisive action is required to jumpstart America's nuclear energy industrial base and ensure our national and economic security by increasing fuel availability and production, securing civil nuclear supply chains, improving the efficiency with which advanced nuclear reactors are licensed, and preparing our workforce to establish America's energy dominance and accelerate our path towards a more secure and independent energy future."

The Reforming Nuclear Reactor Testing at the Department of Energy executive order seeks to maximize the use of the DOE's

authority to approve reactor construction and operation on its sites, which can be used to demonstrate reactor designs independently of the NRC's processes.

The executive order set a target to build and achieve criticality with at least three innovative reactor designs at DOE sites by July 2026. Subsequent shortlisting by the DOE identified 10 companies that it would work with: Aalo Atomics, Antares Nuclear, Atomic Alchemy, Deep Fission, Last Energy, Oklo, Natura Resources, Radiant Industries, Terrestrial Energy, and Valar Atomics.

The executive order Ordering the Reform of the Nuclear Regulatory Commission says:

> "Instead of efficiently promoting safe, abundant nuclear energy, the NRC has instead tried to insulate Americans from the most remote risks without appropriate regard for the severe domestic and geopolitical costs of such risk aversion."

In February 2014 the NRC had said that its most optimistic scenario for awarding design certification for new small reactors was 41 months, assuming they were light water types (PWR or BWR). The executive order obviously intends to spur a significant acceleration of this and other new build licensing activities, although without specifying how.

US government and Westinghouse

In October 2025 the US government announced a $80 billion strategic partnership with Westinghouse Electric Company and its owners Brookfield Asset Management and Cameco Corporation to deploy a fleet of AP1000 and AP300 reactors across the USA. The partnership includes government-arranged financing and facilitated permitting, with the government receiving a 20% participation interest in cash distributions above $7.5 billion. Each two-unit AP1000 project is expected to create or sustain 45,000 manufacturing and engineering jobs.

Big tech investment

Since about 2023 the world's largest technology companies have made a series of investments in nuclear energy in the United States, both through power purchase/offtake agreements and direct investments/development agreements.

- May 2023 – Microsoft signs PPA from Helion's planned fusion plant.
- September 2024 – Microsoft signs 20-year PPA supporting restart of Three Mile Island 1 (see below).
- October 2024 – Google signs deal with Kairos Power to purchase energy from multiple SMRs.
- October 2024 – Amazon invests $500 million in X-energy.
- December 2024 – Switch signs non-binding agreement with Oklo to purchase energy from 12 GWe of Oklo Aurora powerhouse projects through 2044.
- June 2025 – Amazon signs PPA running to 2042 with Talen's Susquehanna nuclear power plant.
- June 2025 – Google signs PPA for 200 MWe from Commonwealth Fusion Systems' planned plant in Chesterfield County, Virginia.
- January 2026 – Meta signs 20-year PPA for 2176 MWe from the Perry and Davis-Besse plants in Ohio; plus the purchase of energy from uprates at those two plants and the Beaver Valley plant in Pennsylvania.
- January 2026 – Meta agrees funding to support deployment of two Terrapower Natrium sodium fast reactors (2x345 MWe) with delivery from 2032, plus the rights for energy from up to six other Natrium units targeted for delivery by 2035.
- January 2026 – Meta agrees to prepay for power and provide funding to advance Oklo's project to develop a 1.2 GWe power campus in Pike County, Ohio.

New reactor licensing

There are three regulatory initiatives that have enhanced the prospects of building new plants. First is the design certification process, second is provision for early site permits (ESPs) and third is the combined construction and operating licence (COL) process ('Part 52' of Title 10 of the Code of Federal Regulations) as an alternative to the 'Part 50' two-step process of construction permit

followed by operating licence. All have some costs shared by the DOE.

A COL application under 10 CFR Part 52 may reference a design certification and/or an ESP. This allows the issues resolved during the design certification and ESP processes to be precluded from reconsideration during the COL licensing process.

Design certification

As part of the effort to increase US generating capacity, the government and industry have worked closely on design certification for advanced Generation III reactors. Design certification by the Nuclear Regulatory Commission (NRC) means that, after a thorough examination of compliance with safety requirements, a generic type of reactor (say, a Westinghouse AP1000) can be built anywhere in the USA, only having to go through site-specific licensing procedures and obtaining a combined construction and operating licence (see below) before construction can begin. In September 2025 the duration of design certifications was revised to 40 years (previously 15 years).

Designs now having US design certification are:

- The Westinghouse AP1000, which is the first Generation III+ reactor to receive certification[c]. It is a scaled-up version of the Westinghouse AP600 which was certified earlier. It has a modular design to reduce construction time to 36 months. Four are in operation in China, as well as two in the USA.

- The GE Hitachi advanced boiling water reactor (ABWR) of 1300-1500 MWe. Several ABWRs are now in operation and under construction in Japan. Some of these have had Toshiba involved in the construction, and more recently it has been Toshiba that promoted the design most strongly in the USA.[d] Both the Toshiba and the GE Hitachi versions needed to have their design certification renewed from 2012. Toshiba withdrew its design certification renewal application in mid-2016.

- GE Hitachi's Economic Simplified BWR (ESBWR) of 1600 MWe gross with passive safety features, developed from the ABWR. GE Hitachi submitted the application in August 2005, design approval was notified in March 2011, and design certification was in September 2014. The first combined construction and

operating licence (COL) with it was awarded for Fermi 3 in May 2015 and the second for North Anna 3 in June 2017.

- The Korean APR-1400 reactor, which is operating in South Korea since 2016 and in the United Arab Emirates since 2020. Korea Hydro & Nuclear Power submitted a design certification application to the NRC in October 2013 and the revised submission was accepted by the NRC in March 2015. The final safety report was published in September 2018 and design certification was given in May 2019.

- An uprated 250 MWt (77 MWe gross) NuScale Power Module design was accepted by the NRC in July 2023 and approved in May 2025. This design is an uprated version of the 160 MWt (50 MWe gross) NuScale Power Module design, which was earlier accepted for review by the NRC in March 2017 and approved in September 2020. A reactor building would contain up to 12 of the 77 MWe NuScale Power Modules. The design comprises a reactor core, pressurizer and two steam generators integrated within a reactor pressure vessel and housed in a compact steel containment vessel. The uprated design is the basic unit of NuScale's VOYGR plants: VOYGR-12 (924 MWe), VOYGR-6 (462 MWe) and VOYGR-4 (308 MWe). The initial plant was planned to be the UAMPS Carbon Free Power Project, a VOYGR-6 plant at the Idaho National Laboratory, which would be owned by Utah Associated Municipal Power Systems (UAMPS).

Reactor designs formerly undergoing US design certification:

- The US Evolutionary Power Reactor (US EPR), an adaptation of Areva's EPR to make the European design consistent with US electricity frequencies. The main development of the type was to be through UniStar Nuclear Energy, but other US proposals also involved it. The application was submitted in December 2007 and the design certification rule was expected after mid-2015, with delays due to the complexity of digital instrumentation and control systems. Areva then delayed the NRC schedule and in March 2015 indefinitely suspended the application. EPRs have been built in Taishan in China, as well as Finland and France. Two units are under construction in the UK at Hinkley Point C.

- The Mitsubishi US-APWR, a 1700 MWe design developed from that for a 1538 MWe reactor planned for Tsuruga in Japan. The application was submitted in December 2007 and certification

was expected to be completed in February 2016, but Mitsubishi delayed the NRC schedule for “several years”. European certification for the almost identical EU-APWR was granted in October 2014. Two US-APWR reactors were proposed in the Luminant-Mitsubishi application for Comanche Peak, but Mitsubishi has withdrawn from this project.

Several designs of small modular reactors (SMRs) are proceeding with pre-application activities towards NRC design certification application, including:

- GE Hitachi Nuclear Energy submitted licensing documentation to the NRC in December 2019 for the BWRX-300. The company said the design "leverages the design and licensing basis of the NRC-certified ESBWR" and that it "represents the simplest, yet most innovative BWR design since GE began developing nuclear reactors in 1955."

- Holtec International announced in November 2020 that it had commenced licensing procedures with the NRC. A demonstration unit of the 160 MWe Holtec SMR-160 PWR (with external steam generator) had been proposed at the Savannah River Site with DOE support. In December 2023 Holtec announced a new plan to build its first two SMR units – using the 300 MWe version of its SMR design, the SMR-300 – at its Palisades nuclear plant in Michigan. Holtec said it plans to file a construction permit application with the NRC by 2026 and has a target commissioning date for the first SMR-300 in the mid-2030s.

- In March 2025, Switzerland-based Deep Atomic entered the pre-application process for design certification of its MK60 design. The MK60 is a PWR with 60 MWe capacity that also provides 60 MW cooling capacity, which can support the thermal management requirements of a large data centre, according to Deep Atomic.

A fuller account of new reactor designs, including those certified but not marketed in the USA, is in the information page on Advanced Nuclear Power Reactors, or for the small modular reactors, in the page on Small Nuclear Power Reactors.

Early site permit

Early site permits facilitate a utility or project developer to engage the NRC on the suitability of a site to host a nuclear power plant in advance of making firm decisions on either reactor design or whether to proceed with nuclear build. If granted, ESPs are valid for 10 to 20 years from the date of issue, with the possibility of extension by a further 10 to 20 years.

The ESP programme was launched in 2001, attracting four applicants: Exelon, Entergy, Dominion and Southern, for Clinton, Grand Gulf, North Anna and Vogtle sites respectively – all sites then with operating nuclear plants but room for more.

In March 2007, Exelon was awarded the first ESP for its Clinton plant in Illinois, after 41 months' processing by the NRC and public review. The NRC then awarded ESPs to Entergy for its Grand Gulf site, Dominion for North Anna, and Southern for Vogtle. No plant type is normally specified with an ESP application, but the site is declared suitable on safety, environmental and related grounds for a new nuclear power plant. The last three of these 2001 ESPs were replaced by COL applications.

In March 2010, Exelon applied for an ESP for its Victoria County, Texas, site and withdrew the COL application for that project. In 2012 it withdrew the ESP application. PSEG Nuclear lodged an application for an ESP for a new reactor at its Salem/Hope Creek site on the Delaware River in New Jersey in May 2010, and this was granted in May 2016.

The seventh ESP application was for small reactors. In May 2016 Tennessee Valley Authority (TVA) submitted an ESP application to the NRC for up to 800 MWe total generating capacity across multiple small reactors at Clinch River. The application was based on a plant parameter envelope encompassing the light-water SMRs currently under development in the USA by BWX Technologies, Holtec, NuScale Power and Westinghouse. It envisaged that the emergency planning zone need extend only to the site boundary. The ESP, supported by the DOE, was issued in December 2019.

In January 2026 Duke Energy applied for an ESP for a site near the 2200 MWe Belews Creek power plant in North Carolina, which is co-fired by gas and coal and is slated for retirement in the late 2030s. Although the ESP process is technology-neutral, Duke said that it was not considering large light water reactors but instead a range of small and advanced reactors.

Site use permits can be awarded by the DOE for its sites. In December 2019 Oklo received a site use permit for an Aurora reactor to be built at Idaho National Laboratory.

Vogtle 3&4

In April 2008, Georgia Power signed an EPC contract with Westinghouse and The Shaw Group (now CB&I) consortium for two 1200 MWe Westinghouse AP1000 reactors which will be licensed and operated by Southern Nuclear Operating Company (SNOC). Both Georgia Power and SNOC are subsidiaries of Southern Company. JSW in Japan sent forged components to Doosan in South Korea for fabrication. The COL was issued by the NRC in February 2012. Construction start (first concrete) was delayed to late 2012, and then to March 2013, after NRC issued a licence amendment allowing use of a higher-strength concrete that permits the company to pour the foundation of the new reactors without making additional modifications to reinforcing steel bar. At that point ten million working hours had been invested on the site. Shaw (now CB&I) agreed with China's State Nuclear Power Technology Corporation (SNPTC) to deploy engineers with experience in building China's AP1000 units to provide technical support. Following early delays, construction of unit 3 started in March 2013 and unit 4 in November. Fluor joined the project as construction manager in January 2016, taking over part of the CB&I role, and in January 2017 Bechtel became involved with the nuclear islands. The units were initially expected online late in 2019 and September 2020. It is a regulated plant, with guaranteed operational cost recovery.

Reactor pressure vessels and steam generators are from Doosan in South Korea.

Georgia Power as 45.7% owner reduced its earlier cost estimate for building its share of the new plant from $6.4 billion to $6.1 billion as a result of being able to recover financing costs from customers during construction, but this increased to $6.2 billion in 2012 due to delays. Over the life of the plant, the utility's customers will save about $1 billion through federal loan guarantees, production tax credits and the early recovery of financing costs in the rate base. The Georgia Public Service Commission in February 2013 approved Georgia Power's costs for the project and said that the project "remains more economically viable than any other [energy] resource, including a natural gas-fired alternative."

The initial cost estimate for the project was $14 billion. Delays to mid-2014 resulted in a cost increase of $381 million but this was offset by lower interest rates than budgeted. When further delays were announced in January 2015, the company said that cost

escalation was about $10 million per month plus financing cost of about $30 million per month. Minority equity in the project is held by Oglethorpe Power (30%), the Municipal Electric Authority of Georgia – MEAG Power (22.7%), and Dalton city (1.6%).

Loan guarantees totalling $3.5 billion were issued to Georgia Power and $3 billion to Oglethorpe Power in 2014. A further $1.8 billion of loan guarantees were issued to three subsidiaries of MEAG Power in June 2015, making a total of $8.3 billion. (Dalton Utilities did not seek a loan guarantee.) In August 2017 Georgia Power, Oglethorpe Power and MEAG sought further loan guarantees to help them complete the project. In September 2017 the DOE announced conditional commitments for further loan guarantees of up to $3.7 billion: $1.67 billion to Georgia Power, $1.6 billion to Oglethorpe Power, and $415 million to three subsidiaries of MEAG Power. (Dalton Utilities again did not apply.) These were granted in March 2019. The DOE said: "Advanced nuclear energy projects like Vogtle are the kind of important energy infrastructure projects that support a reliable and resilient grid, promote economic growth, and strengthen our energy and national security."

Earlier, in mid-April 2017, Westinghouse said that about $1.5 billion was required to complete the construction of both units, though other estimates are higher. In June Toshiba agreed with the owners that its liability under its 2008 parental guarantee would be capped at $3.68 billion for the completion of the Vogtle units. The sum is part of an $8.9 billion provision in Toshiba's accounts announced in mid-May, covering all four US reactors.

In mid-May 2017 Georgia Power announced that from June, Southern Nuclear Operating Company (SNOC) would take over project management to complete the Vogtle units, leaving Westinghouse simply as the vendor, though supporting EPC and licensing as well as providing access to intellectual property. Southern said that productivity at the site had improved significantly in 2017, with the reactors now two-thirds complete. SNOC will also be the operator. The company said it would "take all actions necessary to hold Westinghouse and Toshiba accountable for their financial obligations."

After a review of options and contingencies, at the end of August 2017 Georgia Power, supported by the co-owners, recommended to the state public services commission (PSC) that construction of both units should be completed, this being the most economic choice for customers. The total rate impact of the project remains less than originally estimated, it said. The recommendation was unanimously approved by the PSC in December 2017.

At the same time Georgia Power announced it had contracted with Bechtel to manage daily construction efforts under the direction of SNOC. Bechtel has been involved with the project since January, correlated with “a marked increase in productivity” providing “every indication that we can do a better job than Westinghouse alone as we move forward to complete the project." Vogtle 3&4 would begin commercial operation in November 2021 and November 2022 respectively, under a new construction schedule. These dates were reaffirmed by Southern Company in September 2020, at which point construction of the two units was 87% complete. However, in April 2021 Southern Company said it was targeting a December 2021 in-service date for unit 3, and in May 2021 officials told the Georgia Public Service Commission that the likely commercial start date was January 2022. In-service dates were moved to Q3 2022 and Q2 2023 in October 2021, before being moved again in February 2022, to Q1 2023 and Q4 2023.

In August 2022 the NRC granted authorization to Southern Company to load fuel and begin commissioning activities at Vogtle 3. Southern said it was aiming to carry out fuel loading before the end of October 2022.

In January 2023 Georgia Power notified the US Securities and Exchange Commission that Vogtle 3's initial criticality would be delayed after vibrations in the plant's cooling system were found and an issue with a dripping valve was identified during start-up and pre-operational testing. A month later, Southern Company announced that the vibration issue had been remediated and testing had resumed. However, an unexpected issue with flow rates through reactor coolant pumps delayed the schedule.

Unit 3 was connected to the grid on 1 April 2023, and entered commercial operation in July. Fuel loading at unit 4 began in August 2023. In October 2023 a motor fault was discovered in a reactor coolant pump at unit 4, slightly delaying its commercial operation to March 2024. In February 2024 vibrations in the cooling system similar in nature to those experienced during the construction of unit 3 were observed at unit 4. In March 2024 unit 4 was connected to the grid.

Georgia Power (45.7% owner) said it had invested about $4.3 billion in capital costs in the project to June 2017 and in August 2018 announced that it had revised its forecast for the cost of its 45% share of the project up to $8.4 billion. The total price for the project in November 2021 was estimated to be over $28 billion. In May 2022 this increased to $30.34 billion.

Watts Bar 2

While the focus is on new technology, TVA undertook a detailed feasibility study which led to its decision in 2007 to complete unit 2 of its Watts Bar nuclear power plant in Tennessee. The 1165 MWe (net) reactor was expected to start up in October 2012 and come online in 2013 at a cost of about $2.5 billion, but this schedule slipped substantially, with major budget overrun to $4.7 billion. Construction had been suspended in 1985 when 80% complete and (after parts were cannibalized to reduce that figure to 61%) resumed in October 2007 under a still-valid permit. The construction permit was extended to September 2016, and in October 2015 TVA received a 40-year operating licence from the NRC. Grid connection was early in June and commercial operation commenced in October 2016. Its twin, unit 1, started operation in 1996.

Completing Watts Bar 2 utilized an existing asset, thus saving time and cost relative to alternatives for new base-load capacity. It was expected to provide power at 4.4 ¢/kWh, 20-25% less than coal-fired or new nuclear alternatives and 43% less than natural gas. It is a regulated plant, with guaranteed cost recovery.

In 2014, before start-up, TVA ordered four new steam generators for the unit to be installed in 2022 at a cost of $160 million. The early 1980s ones were made of an alloy that is prone to stress corrosion cracking. Those in unit 1 were replaced after nine years of operation, and the vast majority of US PWRs have had replacements.

Nuclear reactor restarts

Palisades

In September 2017 Entergy announced that it would keep its Palisades nuclear plant in Michigan open until 2022. The company had previously announced in December 2016 that it planned to close the 789 MWe net unit in October 2018 due to economic factors in the partly deregulated market. The reactor was shut down in May 2022 and sold to Holtec International in June for decommissioning.

In light of the DOE's publication of its Civil Nuclear Credit Program – aiming to keep marginal units in deregulated environments online to help accelerate the US energy transition – in September 2022, Holtec international began exploring the possibility of restarting the plant. In November 2022 the DOE rejected Holtec's application that sought funding under the Civil Nuclear Credit Program to reactivate Palisades.

The following month, Holtec announced plans to launch a second attempt to secure federal funding to restart the unit. In January 2023 the Board of Commissioners of Allegan County, Michigan voted unanimously in favour of Holtec's bid to obtain federal funding to restart the unit.

In March 2023 Holtec applied for federal funding from the DOE under the Civil Nuclear Credit Program to restart the Palisades plant, which it believes would cost more than $1 billion. In September 2023 a long-term power purchase agreement was agreed between Palisades Energy and Wolverine Power Cooperative. Later that month, Holtec applied to the NRC for reauthorization of power operations at the plant. Also in the same month, Wolverine Power Supply submitted an application for funding through the US Department of Agriculture's Empowering Rural America (New ERA) $9.7 billion grant and loan initiative that is funded by the Inflation Reduction Act.

In October 2025 new fuel was delivered to the plant. The restart was initially expected by the end of 2025, but was delayed by steam generator tube repairs, with the restart now expected in 2026.

Crane Clean Energy Center (Three Mile Island 1)

Three Mile Island 1 was shut down in September 2019 due to economic challenges (see Electricity Market Challenges, below). Although the unit had been licensed to operate until 2034, Exelon had announced in May 2017 that it would be closed if policy reforms recognising nuclear as a low-carbon electricity producer were not enacted.

The plant's name was changed to the Crane Clean Energy Centre in 2025. Constellation has signed a 20-year power purchase agreement with Microsoft and intends to bring the unit back into operation in 2028. In September 2025 Constellation said the project was ahead of schedule with commercial operation now expected in 2027.

In November 2025 the DOE issued a $1 billion loan to Constellation to support the restart.

Duane Arnold

The single-unit Duane Arnold plant was licenced to operate until 2034 but owner NextEra opted to close it in 2020 after a major power purchase customer shortened its contract and the plant suffered non-nuclear damage in a major storm. Fuel was removed and placed in dry storage and the plant entered a non-operational status. Duane Arnold's grid interconnection rights were split up to support a number of prospective solar projects.

However, by 2024 regional increases in demand for continuous clean power caused NextEra to reassess the unit's economics and begin work to bring it back into operation. In January 2025 NextEra applied to the NRC to restore its operational status. In September 2025 the Federal Energy Regulatory Commission agreed to reconsolidate the unit's previous interconnection rights. In October 2025 Google signed a 25-year power purchase agreement with NextEra for the plant. In January 2026 Linn County approved rezoning to support the restart, which is expected in early 2029.

Nuclear power reactors restarting

Site	MWe net	Proponent/utility	Licensing status	Loan guarantee; restart date	Licence expiration, once restored
Palisades	789	Holtec	January 2023: Request filed to restore operational status July 2025: Authorized to receive fresh nuclear fuel	$1.52 billion; 2026	March 2031
Crane Clean Energy Center (Three Mile Island 1)	835	Constellation	November 2024: Request filed to restore operational status	$1 billion; 2027	April 2034

Site	MWe net	Proponent/utility	Licensing status	Loan guarantee; restart date	Licence expiration, once restored
Duane Arnold	615	NextEra	January 2025: Request filed to restore operational status	early 2029	February 2034

Subtotal restarting: 2239 MWe

New reactor plans and proposals

Proposed nuclear reactors

Kemmerer 1

In March 2024, TerraPower submitted a construction permit application for a Natrium reactor at Kemmerer in Lincoln County, Wyoming, near PacifiCorp's Naughton coal-fired plant. The construction permit was authorized in March 2026.

The Natrium reactor is based on the GE Hitachi (now GE Vernova Hitachi Nuclear Energy) 345 MWe (840 MWt) PRISM sodium-cooled fast reactor coupled with a molten salt-based energy storage system. The Natrium storage technology allows the system's output to be increased to 500 MWe for five-and-a-half hours.

Terrapower is also constructing a 'sodium test and fill facility' nearby to test components for use in the Natrium plant as well as to carry out the initial transfer of sodium to the plant. Construction of this began in June 2024.

TerraPower was one of two reactor developers (the other being X-energy) to receive the first cost-shared awards of $80 million each under the DOE's Advanced Reactor Demonstration Program (ARDP) in October 2020. The aim of the ARDP is to develop, build and operate two advanced reactors within seven years. The DOE said it would invest up to $3.2 billion over this period in the ARDP.

TerraPower announced in October 2022 that construction of a Natrium fuel fabrication facility would commence in 2023 at the

Global Nuclear Fuel - Americas site in Wilmington, North Carolina. However, following Russia's invasion of Ukraine, it has not been viable to source high-assay low-enriched uranium (HALEU) from Russia. The Natrium reactor uses metallic HALEU fuel.

In December 2025 the NRC completed its final safety evaluation for TerraPower's construction permit application, one month ahead of schedule. A construction licence was authorized in March 2026, at which time Terrapower said it would begin construction within weeks. The company had previously said it expected to submit an operating licence application for the Kemmerer plant in 2026. Pre-licensing activities have been underway since mid-2021.

Long Mott Energy

A construction permit application for a four-module (4 x 80 MWe) Xe-100 reactor plant at the Long Mott Generating Station in Calhoun County, Texas was submitted in March 2025.

Xe-100 designer X-energy (along with TerraPower, see above) received a cost-shared award of $80 million in October 2020 under the DOE's Advanced Reactor Demonstration Program (ARDP).

The Xe-100 is a helium-cooled pebble-bed HTR of 200 MWt, 80 MWe would use TRISO-coated uranium oxycarbide fuel made from high-assay low-enriched uranium (HALEU). Factory-made units with a 60-year operating lifetime would be transported to the site by road and installed.

X-energy opened a TRISO pilot fuel fabrication facility at Oak Ridge National Laboratory, Tennessee in 2016 and in April 2022 its TRISO-X subsidiary submitted a Category II special nuclear material licence application the TRISO-X Fuel Fabrication Facility, also at Oak Ridge. The NRC expects the licensing review to last until mid-2026. In the meantime, in October 2022, TRISO-X subsidiary commenced construction work on the facility.

The Long Mott facility would provide process heat and electricity to The Dow Chemical Company's Seadrift Operations site. Long Mott Energy is a wholly-owned subsidiary of Dow.

Clinch River

In April 2025, Tennessee Valley Authority (TVA) applied for a construction permit for a BWRX-300 at its Clinch River site in Oak

Ridge, Tennessee. The NRC accepted the application for review in July 2025 and expects to complete its 17-month review by December 2026. An early site permit (ESP) for at least two SMRs (up to 800 MWe) at the site was issued in December 2019.

The ESP application did not reference a particular technology, but in February 2013 Babcock & Wilcox signed an agreement with TVA to build up to four 180 MWe mPower units at Clinch River, with design certification application intended to be submitted to the NRC in 2015.

Energy Northwest, Hanford

Energy Northwest commenced pre-application activities in August 2024 for a construction permit application for up to 12 X-energy Xe-100 units on the Department of Energy's Hanford site, adjacent to Columbia Generating Station in Benton County, Washington. Energy Northwest has selected AtkinsRéalis as technical advisor for licensing and design activities.

In October 2024 X-energy announced $500 million financing from Amazon's Climate Pledge Fund and Citadel Founder and CEO Ken Griffin, Ares Management, NGP, and the University of Michigan. Amazon and X-energy are developing four units (320 MWe), with the option to increase that project to 12 units and 960 MWe. The facility will be known as the Cascade Advanced Energy Facility.

Palisades SMR-300

Holtec plans to build a 10 GWe fleet of SMR-300 reactors in North America working with Hyundai Engineering & Construction, with the first two deployed at its Palisades site. Under its 'Mission 2030' programme announced in February 2025, Holtec plans to have commenced operation of the pair of SMR-300 units as well as restarted the 789 MWe net PWR at Palisades by the end of 2030. Earlier pre-application activities towards a construction licence application on its SMR-160 design were suspended in December 2023. In December 2025 Holtec submitted a partial construction permit application to the NRC for the two SMR-300 units at Palisades.

In December 2025 the US Department of Energy selected Tennessee Valley Authority and Holtec International to each receive $400 million in federal cost-shared funding to support early

deployments of advanced light-water small modular reactors. The funding supports TVA's Clinch River BWRX-300 project and Holtec's planned SMR-300 units at Palisades.

INL Aurora

Oklo aims to build its first Aurora Powerhouse reactor at Idaho National Laboratory for which the DOE has issued a site use permit. In March 2025, Oklo announced that it had increased the capacity of the Aurora reactor 'platform' from 50 MWe to 75 MWe, offering a flexible output range of 15-75 MWe. The sodium-cooled fast neutron reactor would use high-assay low-enriched U-Zr metallic fuel.

Oklo states that the design "is informed by a robust foundation of prior US Department of Energy (DOE) demonstrations and operational experience with fast reactor technologies, most notably the Experimental Breeder Reactor-II (EBR-II) and the Fast Flux Test Facility (FFTF)."

Oklo was given the go-ahead to begin site characterization work at INL in November 2024 for its first Aurora unit, and in September 2025 held a groundbreaking ceremony. Earlier, in August 2025 Oklo announced that Kiewit Nuclear Solutions Co will serve as lead constructor supporting the design, procurement, and construction of the Aurora Powerhouse.

In November 2025 the DOE approved the preliminary documented safety analysis for Oklo's Aurora Fuel Fabrication Facility at INL, allowing assembly to begin. Commercial operation is targeted for late 2027 to early 2028.

Oklo previously submitted a combined construction and operating licence (COL) application to the NRC in March 2020 to build and operate a 1.5 MWe Aurora fast heatpipe reactor at the INL site. The NRC accepted this application in June 2020 but denied the COL ‘without prejudice’ in January 2022, saying that important information was missing from the application.

The company recommenced pre-licensing activities with the NRC later that year and in July 2025 announced it had completed a pre-application readiness assessment for Phase 1 of a revised COL application based on the 75 MWe design.

Fermi America

In June 2025 Fermi America submitted the first part of its COL application to build four AP1000 reactors in Carson County, Texas, at the President Donald J Trump Advanced Energy and Intelligence Campus, also referred to as Project Matador. The second part of the COL application was submitted in August. The company aims to put the first reactor in operation in 2032, followed by the others in 2034, 2035 and 2036. The NRC accepted the first two parts of the application in September 2025. Fermi America signed agreements with Hyundai Engineering & Construction and other Korean firms to support design and construction of the project.

The NRC accepted the first two parts of the application in September 2025. Fermi America signed agreements with Hyundai Engineering & Construction and other Korean firms to support design and construction of the project.

Hermes test reactor

In December 2023 the NRC issued a construction permit to Kairos Power for the Hermes test reactor, which supports the development of the KP-FHR fluoride salt-cooled high-temperature reactor.

Construction activities at the Oak Ridge National Laboratory site in Tennessee started in July 2024. Nuclear safety-related construction began in May 2025 with the pouring of concrete for foundation piers. Operation is targeted for 2027.

In November 2024 construction permits were issued for the Hermes 2, which comprises two 35 MWt reactors. Unlike the Hermes reactor, the two Hermes 2 reactors will produce electrical power. Each Hermes 2 unit will use an intermediate salt (BeNaF) loop to exchange heat from the primary (FLiBe) salt to the common power generation system.

The 35 MWt reactor will use tri-structural isotropic (TRISO) high assay low enriched uranium (HALEU) fuel particles embedded in a carbon matrix pebble. In January 2026 Kairos finalized a contract with the DOE to receive the requisite HALEU.

Kewaunee

In September 2025 EnergySolutions and WEC Energy announced they were exploring the construction of new nuclear capacity at the Kewaunee nuclear power plant site in Wisconsin. In January 2026 EnergySolutions submitted a notice of intent for a licensing submission regarding Kewaunee, saying it was evaluating applications for an ESP, construction permit or combined licence. EnergySolutions intends to make its submission by June 2028.

A 556 MWe PWR began operation at the site in 1974 and was shut down in 2013 with its owner, Dominion Energy, citing poor economic conditions. Following its shutdown, ownership was transferred to EnergySolutions to decommission the reactor.

Site	Technology	MWe gross	Proponent/utility	Licensing status
Kemmerer 1, Wyoming	Natrium	345	US SFR Owner (a TerraPower subsidiary)	Construction permit application March 2024; issued March 2026.
Seadrift, Texas	Xe-100	4x80	X-energy, Long Mott Energy (a Dow subsidiary)	Construction permit application March 2025.
Clinch River, Tennessee	BWRX-300	1 x 300	TVA	ESP application May 2016, issued December 2019. Construction permit application April 2025.
Adjacent to Columbia Generating Station, Washington	Xe-100	12 x 80	X-energy, Energy Northwest	
Palisades, Michigan	SMR-300	2 x 300	Holtec	Partial construction permit application submitted December 2025.
Idaho National Laboratory, Idaho	Aurora	15	Oklo	DOE issued site use permit in 2019. In July 2025 Oklo completed a pre-application readiness assessment for Phase 1 of a COL application.
Fermi America 1-4, Carson County, Texas	AP1000	4 x 1250	Fermi America	Partial COL application submitted June 2025.

Subtotal proposed: 21 small (2540 MWe) and 4 large (5000 MWe)

Other proposals, suspended or cancelled

Summer 2&3

In May 2008, South Carolina Electricity & Gas (SCANA subsidiary) and state-owned Santee Cooper signed an EPC contract with Westinghouse and the Shaw Group (now CB&I) consortium for two 1200 MWe Westinghouse AP1000 reactors. The total forecast cost of $9.8 billion included inflation and owners' costs for site preparation, contingencies and project financing, though the last was reduced and the total estimated in April 2012 was $9.2 billion. In October 2014 the cost was estimated at over $11 billion, and in 2015 SCEG amended the EPC contract to choose a fixed price option for completion of the units. In November 2016 the state public service commission agreed for SCEG's 55% share to be $7.66 billion, excluding financing, with the company's return on equity reduced to 10.25%. "These delays and related cost increases are principally due to design and fabrication issues associated with the production of submodules used in construction of the units," according to SCANA. Fluor joined the project as construction manager in January 2016, taking over the CB&I role. In February 2017 the anticipated completion dates for the two units were April 2020 and December 2020.

The COL was issued by the NRC at the end of March 2012, and construction of unit 2 commenced in March 2013, with first main concrete. That for unit 3 was in November 2013. (In September 2011 SCEG had started to assemble the containment vessel for the first unit – 43 mm thick, from Chicago Bridge & Iron – and was starting construction on the four low-profile forced-draft cooling towers.) Reactor pressure vessels and steam generators are from Doosan in South Korea. A crane capable of lifting 6800 tonnes is installed onsite, though the heaviest component was 1550 tonnes. SCEG's loan guarantee application was accepted by the DOE and the project was short-listed in May 2009, though nothing has happened since then. It is a regulated plant, with guaranteed operational cost recovery.

In 2014 it was announced that SCEG's stake in the project would be increased to 60% by acquisition of 5% from Santee Cooper after the plant starts up, for about $500 million, leaving it with 40%. Duke Energy Carolinas had been seeking up to 10% of the project from Santee Cooper, but this plan was dropped in January 2014.

Following Westinghouse filing for Chapter 11 protection from creditors in March 2017, SCANA reviewed the project and initially

expected resources from Westinghouse and Toshiba – including a so-called parental guarantee from Toshiba – to be adequate to compensate for the additional costs. These, together with a surety bond and an escrow of AP1000 intellectual property and software, were considered. SCANA and Santee Cooper had intended to take over project management to complete the Summer units, leaving Westinghouse simply as vendor, though supporting EPC and licensing as well as providing access to intellectual property, as with Vogtle. In mid-April Westinghouse told SCANA that about $1.5 billion was required to complete construction of both units – $829 million more than it was entitled to charge under the EPC contract, but less than the liability amount for it and Toshiba for breach of EPC contract. SCE&G and Santee Cooper reached agreement with Westinghouse and Toshiba to settle for $2.168 billion. Of this $1.192 billion will go to SCE&G for its 55% ownership of the project, with $976 million to Santee Cooper, which owns 45%. Analysis of detailed schedule and cost data provided by Westinghouse and EPC subcontractor Fluor showed unit 2 would not be completed until December 2022 and unit 3 not before March 2024 – four years after the most recent completion date provided by Westinghouse. The overall project was 64.1% complete at the end of March 2017, and "about two-thirds" complete in July.

At the end of July Santee Cooper decided to halt construction in the light of “significant challenges” in completing the two reactors, notably uncertain costs, the uncertain availability of production tax credits, and reduced demand forecast. Also "the current political landscape has reduced the urgency for emissions-free base-load generation." It found that completing the project would cost the company $8 billion plus about $3.4 billion in interest, with schedule delays contributing to the increased interest. It had already spent $4.7 billion on construction and interest to date for its 45% share of the project. SCE&G had been evaluating options, including completion of only one unit, but concluded that completion of both units would be “prohibitively expensive” – about $9.9 billion for its 55% share of the project. SCANA said that completing only unit 2 would have resulted in a combined cost that was less than that previously approved by the South Carolina Public Services Commission under the fixed price option for completing the two nuclear units, but Santee Cooper’s decision ruled this out. “Ceasing work on the project was our least desired option, but this is the right thing to do at this time," and would accordingly apply to the state public services commission to permit this and allow it to recover from ratepayers about $4.9 billion it has spent.

Santee Cooper said that during the project wind-down it will continue to investigate the potential for federal support or "additional partners" that might make the project economic, and SCE&G echoed this. The state government then considered trying to sell Santee Cooper or take other action to revive the project, and SCE&G said in mid-August that it would withdraw its petition to the state public services commission, to allow for possible new partners. Duke Energy said it was not interested.

Westinghouse said: "The South Carolina economy is sure to feel the negative impact of losing over 5000 high-paying, long-term jobs, as well as not having available the reliable, clean, safe and affordable energy these units would provide. Also, at a time when other nuclear plants are being retired, the US energy sector is sure to feel the stunting impact of walking away from these two nuclear units."

In September 2017 the state governor released a report written 18 months earlier by Bechtel, highlighting eight significant contractual and management problems that required resolution*. The report detailed numerous recommendations, but suggested that the most important step for the consortium was to create a new "more achievable" project schedule.

Later in September 2017, SCANA and its subsidiaries received a federal subpoena for a broad range of documents related to the Summer plant expansion.

* The report found that some issues were to be expected due to the choice of reactor type – the project was due to be the first AP1000 reactor built in the USA – and the preceding hiatus in nuclear new build activity in the country. However it also highlighted the following eight significant contractual and management problems that required resolution:

- While the consortium's engineering, procurement and construction plans and schedules are integrated, the plans and schedules are not reflective of actual project circumstances.
- The consortium lacks the project management integration needed for a successful project outcome.
- There is a lack of a planned vision, goals and accountability between the owners and the consortium.
- The contract does not appear to be serving the owners or the consortium particularly well.

- The detailed engineering design is not yet completed, which will subsequently affect the performance of procurement and construction.
- The issued design is often not constructible, resulting in a significant number of changes and causing delays.
- The oversight approach taken by the owners does not allow for real-time, appropriate cost and schedule mitigation.
- The relationship between the consortium partners (Westinghouse Electric Company and Chicago Bridge & Iron) is strained, caused to a large extent by commercial issues.

In September 2020 Santee Cooper and Westinghouse finalized the terms of a settlement over ownership of equipment associated with the VC Summer plant. Earlier in May 2019, Santee Cooper had asked a New York bankruptcy court to dismiss Westinghouse's claim of ownership of the same equipment. The two companies eventually agreed to split the net sales proceeds for major non-installed nuclear equipment. For major installed nuclear equipment, Santee Cooper was to receive 90% and Westinghouse 10%. For other equipment that could be used in other nuclear projects, 67% of the sale proceeds were to go to Santee Cooper and 33% to Westinghouse. Santee Cooper has 100% ownership of the remaining project equipment. Westinghouse was to have responsibility for marketing the nuclear equipment.

However, Brookfield Asset Management has been in talks to acquire and restart the two units at VC Summer. In October 2025 Santee Cooper's board approved Brookfield's proposal, granting the two parties an exclusive negotiation window expected to last several weeks as they finalize terms. In December 2025 the parties signed a memorandum of understanding (MoU) under which Santee Cooper would receive $2.7 billion if a final investment decision is reached. The MoU sets a deadline of June 2026 for Brookfield to determine initial feasibility.

Bellefonte

Tennessee Valley Authority had a pair of uncompleted 1213 MWe PWR reactors: Bellefonte 1&2. Construction on these units was abandoned in 1988 after $2.5 billion had been spent and unit 1 largely (88%) completed and unit 2 about 58% completed. In February 2009, the NRC reinstated the construction permits for these (and later the status of the reactors classified as 'deferred').

Today unit 1 is considered no more than 55% complete due to the transfer or sale of many components and the need to upgrade or replace others, such as the instrumentation and control systems, reactor pressure vessel, steam generators and main condenser tubing. In August 2011 TVA opted to complete unit 1 at a cost of about $4.9 billion rather than building a new AP1000 reactor as unit 3* (see Appendix 3: COL Applications). TVA then asked the NRC in 2011 to defer consideration of its COL for units 3&4 (AP1000 option), and in February 2016 it withdrew the COL application.

* In August 2010, TVA had committed to spending $248 million in the year to September 2011 towards work at Bellefonte[8] and an engineering contract was awarded to Areva SA in October 2010 for work on unit 1, including engineering, licensing and procurement of long-lead materials in support of a possible start-up date in the 2018-19 timeframe. Following TVA's 2011 decision to proceed, the Areva contract included construction and component replacement work on the plant's nuclear systems, a digital instrumentation and control (I&C) system, a modernized control room, a plant simulator for personnel training plus fuel design and fabrication. Areva contracts amounted to some $1 billion, with heavy construction to start when Watts Bar 2 was completed. In late 2013 TVA revised the estimated cost to $7.4 to $8.7 billion.

However, TVA's 20-year integrated resource plan in 2015 did not have Bellefonte 1&2 as a firm prospect, and it projected 2028 completion of unit 1 as having the effect of increasing system costs. Later in 2015 the company said it would defer consideration of completing unit 1 for a decade. In May 2016 the TVA board decided to offer the plant for sale at auction, and in November Nuclear Development LLC agreed to buy it for $111 million.

Nuclear Development said it intended to invest up to $13 billion from 2017 to complete the plant, and it was lobbying for a $5 billion loan guarantee. Bellefonte is a regulated plant, with guaranteed cost recovery. In mid-2018 the company signed an agreement with SNC-Lavalin to finish building the plant once the purchase is completed. Completion of unit 1 was then anticipated in 2024. In November 2018 Nuclear Development applied to the NRC to transfer the construction permits and announced its intention to involve Framatome in the project, but late in 2019 the NRC had not yet undertaken a review of the application. The sale is contingent upon NRC approval, and the company said that construction depends both on a loan guarantee (it was seeking $8.6 billion) and securing power purchase agreements.

Lee

Duke Energy lodged a COL application in December 2007 for two Westinghouse AP1000 units for its William States Lee III plant at a new site near Charlotte in Cherokee County, South Carolina, to provide power for North Carolina. The company was seeking a loan guarantee and was considering regional partnerships to build the plant. The environmental review for NRC was completed in December 2013, showing no problems, the safety evaluation review was completed in August 2016 and the COLs issued in December 2016. Duke told NRC in 2012 that it was revising its COL application to move the nuclear island of both Lee units by some 20 metres to make excavation and construction easier. Duke had spent $471 million on licensing, planning and pre-construction activities for the plant to February 2016. If proceeding, the 1117 MWe (net) units were then expected online in 2024 and 2026. In August 2017 the company announced: "The risks and uncertainties to initiating construction on the Lee nuclear project have become too great, and cancellation of the project is the best option for customers."

In October 2025 Duke Energy filed its 2025 Carolinas Resource Plan, which includes the possibility of deploying large LWR capacity at the Lee site.

Turkey Point 6&7

NextEra Energy subsidiary Florida Power & Light (FPL) applied in June 2009 for a COL for two Westinghouse AP1000 reactors at Turkey Point in Florida where two 693 MWe PWR units (3&4) are operating and were uprated in 2012-13. (Unit 5 is a 1190 MWe combined cycle gas plant; units 1&2 are 400 MWe oil/gas units.) In 2011 the Florida Public Service Commission approved a levy towards construction of the reactors, and in May 2014 the state government approved the project, with new transmission lines.

The company said in April 2014 that it expected to start operation of the first new unit in June 2022 and the second a year later, but in January 2015 changed this to 2027 and 2028, due to “NRC licensing schedule adjustments and changes to the Florida nuclear cost recovery law,“ which delay the start of site works. The COL was approved by the NRC in April 2018.

South Texas Project 3&4

Units 3&4 at South Texas Project (STP) were envisaged as a merchant plant with two 1356 MWe Advanced Boiling Water Reactors (ABWR)[j]. The COL application was submitted in September 2007 by site operator STP Nuclear Operating Company

(STPNOC) on behalf of the project owner, which was then a 50:50 partnership between NRG Energy and the City Public Service Board (CPS Energy) of San Antonio. Ownership of STP units 1&2 (Westinghouse PWRs) is Constellation Energy (44%) – which purchased NRG Energy's share in November 2023 – CPS Energy (40%) and Austin Energy (16%).

In March 2008, NRG with Toshiba subsidiary Toshiba America Nuclear Energy (TANE) formed Nuclear Innovation North America (NINA – 88% NRG; 12% TANE) to develop the project. In February 2009, TANE entered into an engineering, procurement and construction (EPC) agreement that would convert into a turnkey contract once the final decision to proceed with the project had been taken. Following TANE's later announcement that the project would cost $4 billion more than the $13 billion that was previously estimated, in February 2010 CPS Energy decided to reduce its stake to 7.625%, with NINA increasing its share to 92.375%.

In May 2010, Japanese utility Tokyo Electric Power Company (Tepco), which had been acting as technical consultant to the project, agreed to take 10% of NINA's stake for $155 million, with an option to later double its holding. The deal was conditional on a DOE loan guarantee being awarded to the project. However, in April 2011, based largely on low natural gas prices in Texas compounded by the March 2011 accident at Tepco's Fukushima Daiichi plant in Japan, NRG decided to pull out of the project and write off its $331 million investment in it. Toshiba had spent $150 million and persevered with the project, though it wrote off $305 million (JPY 31 billion) on NINA in 2014. NINA was dissolved in 2018.

COLs for each of the two units were issued in February 2016.[k] However, Toshiba's withdrawal of the application for design certification renewal in mid-2016 effectively put the project on hold. In May 2018, Toshiba announced its withdrawal from the project, stating that it was no longer financially viable. Toshiba said its decision to exit the project was in line with its policy "to eliminate risk from the overseas nuclear power business, particularly from construction-related cost overruns in nuclear power plant construction projects." Toshiba stated it had sought, but failed to find investors to participate in the project.

UAMPS

The UAMPS Carbon Free Power Project, a six-module NuScale SMR plant at the Idaho National Laboratory, would be owned by Utah Associated Municipal Power Systems (UAMPS) that comprises 48

members from Utah, California, Idaho, Nevada, New Mexico and Wyoming. UAMPS planned to submit a COL application by January 2024. In 2013 NuScale secured up to $226 million DOE support for the original 45 MWe design. The DOE granted permission to site the plant on the 2300 square km Idaho National Laboratory estate, reportedly in the southern part of it. Under this agreement UAMPS had ten years to begin operating the first module, and this would trigger a 99-year lease for the plant.

In October 2020 the DOE approved a $1.335 billion cost-share award, allocated over 10 years, to a special purpose entity wholly-owned by UAMPS – the Carbon Free Power Project – for the development and construction of the planned six-module plant (then 60 MWe per module). The award represented around one-quarter of the development and construction costs over ten years. Projected LCOE was about $55/MWh. In November 2020, the module power was uprated to 77 MWe, lowering the overnight capital cost from $3600/kWe to $2850/kWe, according to NuScale.

However, a UAMPS meeting held in October 2022 indicated significantly higher costs for the project than first estimated. Inflationary pressures, such as the rising price of steel could push the power cost from $55/MWh to between $90 and $100 per MWh.

In November 2023 UAMPS announced that it had mutually agreed to cancel the CFPP due to the inability to reach the 80% subscription rate required to support the development.

Fermi 3

This is a reference unit for GE Hitachi's ESBWR design, proposed by Detroit Edison in Michigan, but the company has not yet committed to proceeding. A COL application was made in 2008 and environmental approval was received in January 2013. Full design certification of the ESBWR in 2014 allowed the safety evaluation to proceed, and the COL was approved in May 2015.

Levy County, Florida

Site works started for two 1200 MWe Westinghouse AP1000 reactors on a greenfield site in Florida, and to January 2012 some $860 million had been spent on this. The company expected to have spent about $1 billion on the design, acquisition of heavy equipment and site works by the time it secures NRC approval. In September 2008, Progress Energy Florida signed an EPC contract with Westinghouse and The Shaw Group (now CB&I) consortium.

The contract is for $7.65 billion ($3462/kWe), of an overall project cost of about $14 billion.

In August 2013 Duke Energy resolved to terminate the 2008 EPC contract as "a result of delays by the NRC in issuing COLs for new nuclear plants, as well as increased uncertainty in cost recovery caused by recent legislative changes in Florida." It continued to pursue the COLs in order to keep the option open. In April 2014 Duke announced plans to build 2745 MWe of gas-fired capacity by 2021 instead of proceeding with the Levy County nuclear plant in the original timeframe. Duke Energy Florida was planning to sell all the long-lead time equipment it had ordered by the end of 2014, but it was in dispute with Westinghouse over EPC contract termination. In October the Florida Public Service Commission ordered Duke to repay to ratepayers $54 million it had collected in advance to fund the 'cancelled' project. In October 2016 the NRC approved the COLs.

The last estimated operational dates were 2024-25, the delay being due to "lower-than-projected customer demand, the lingering economic slowdown, uncertainty regarding potential carbon regulation and current low natural gas prices." The revised cost was $19-24 billion. It would be a regulated plant, with guaranteed cost recovery. In August 2017 Duke Energy cancelled the project, citing the Westinghouse bankruptcy and slowing energy demand, and said it would not maintain the licences.

North Anna 3

In December 2010, Dominion announced that it had agreed with Mitsubishi Heavy Industries to build a US-APWR unit, but in April 2013 Dominion announced that it had reverted to the ESBWR as preferred technology (as originally selected in 2005), and would amend its COL application accordingly. The COL for the ESBWR was issued in June 2017. Dominion quotes 1453 MWe net (summer capacity) for the unit there. In May 2013 it agreed a construction contract with GE Hitachi and Fluor, conditional upon proceeding. Dominion said it will make a decision on building in due course, and hence it remains as 'proposed' according to the World Nuclear Association. Dominion suggests start-up in 2028 if it proceeds. It had spent $345 million on the project to early 2016. It is a regulated plant, with guaranteed cost recovery. A consultant to the state has estimated the cost of the plant as $19.3 billion including financing, or $13,283/kW, and Dominion has said that such a figure would not be unreasonable.

Harris 2&3

Progress Energy lodged a COL application for two AP1000 units at its Shearon Harris site at New Hill in North Carolina in February 2008. This was proceeding towards being granted at the end of 2014. Expansion of the plant would require raising the water level of Harris Lake by 6 metres, and relying on the Cape Fear River as backup cooling water. However, in May 2013 Duke Energy (which had taken over Progress) asked NRC to suspend the COL review due to projected electricity demand being low for next 15 years.

Comanche Peak

Vistra subsidiary Luminant planned to build two US-APWR units as units 3&4 at Comanche Peak in Texas. In May 2011 the NRC concluded that there were no environmental considerations that would hinder the project. Luminant's loan guarantee application was accepted by DOE and it was understood that this was the first alternative to the four shortlisted projects, two of which are now not proceeding for the time being. The application for design certification was submitted in December 2007 and certification was expected to be completed in February 2016, but Mitsubishi delayed the NRC schedule for “several years”. Meanwhile Mitsubishi has withdrew as a joint venture partner.

Calvert Cliffs 3

Unistar, now owned by EdF, planned to build a 1710 MWe Areva US-EPR alongside Constellation's units 1&2, as a merchant plant. Exelon, merging with Constellation (in which EdF has 49.9% equity) said in November 2011 that with the advent of shale gas, a new nuclear plant at Calvert Cliffs was "utterly uneconomic" by a factor of about two.

The design certification application was submitted in December 2007 and the design certification rule was expected after mid-2015, with delays due to the complexity of digital instrumentation and control systems. Areva then delayed the NRC schedule and in March 2015 indefinitely suspended the application.

Salem 3

PSEG was issued in May 2016 with an early site permit for up to two new Salem reactors at Hope Creek, NJ. No reactor technology was specified.

Other planned or proposed new US nuclear capacity is described more fully in Appendix 3: COL Applications.

Site	Technology	MWe gross	Proponent/utility	Licensing status	Status
Virgil C. Summer 2&3, SC	AP1000	2 x 1250	South Carolina Electricity & Gas	COL lodged 27/3/08, issued March 2012, terminated March 2019	Construc commen March 2((unit 2) a Novemb 2013 (un 3). Construc abandon July 2017
Turkey Point 6&7, FL	AP1000	2 x 1250	Florida Power & Light	COL lodged 30/6/09, COL issued April 2017	
Fermi 3, MI	ESBWR	1600	Detroit Edison	18/9/08, COL issued May 2015	No decis to procee
North Anna 3*, VA	ESBWR[f]	~1500	Dominion	20/11/07, COL issued June 2017, ESP issued	On hold from Sep 2017
Bellefonte 1&2[g, h], AL	B&W PWR (partly built)	2 x 1263	Nuclear Development LLC (sale pending from Tennessee Valley Authority)	30/10/07 for units 3&4[h] but COL withdrawn 2016	Seeking loan guarante
Salem 3/Hope Creek, NJ	unspecified	1200?	PSEG Nuclear	ESP issued May 2016	
Victoria County, TX	ESBWR	3200	Exelon (merchant plant)	03/9/08 but withdrawn, ESP application 25/3/10, but withdrawn Oct 2012	
Piketon (DOE site leased to USEC), OH	US EPR	1710	Duke Energy		
Payette county, ID	APWR	1700	Alternate Energy Holdings Inc. (merchant plant)	Plans stalled since 2012	
Fresno, Ca	US EPR	1710	Fresno Nuclear Energy Group		
Amarillo, TX	US EPR	2 x 1750	Amarillo Power (merchant plant)		

Site	Technology	MWe gross	Proponent/utility	Licensing status	Status
Levy County, FL	AP1000	2 x 1250	Duke Energy (formerly Progress Energy)	30/07/08, COLs approved Oct 2016 and cancelled April 2018	Project cancelle Aug 201
Callaway[i], MO	Westinghouse SMR	5 x 225	Ameren Missouri	24/07/08 for EPR, then withdrawn; SMR proposal suspended	
Shearon Harris 2&3, NC	AP1000	2 x 1250	Duke Energy (formerly Progress Energy)	19/02/08, COL suspended May 2013	
Grand Gulf, MS	ESBWR[i]	1600	Entergy	27/02/2008, COL application withdrawn 9/15, ESP issued	
Comanche Peak, TX	US-APWR	2 x 1700	Luminant (merchant plant)	19/09/08, COL suspended 11/13	
Bell Bend (near Susquehanna), PA	US EPR	1710	PPL/Talen (merchant plant)	10/10/08, COL review suspended 2014 but EIS approved. COL application withdrawn Aug 2016	Suspend indefinite
Calvert Cliffs[*], MD	US EPR	1710	UniStar Nuclear (merchant plant)	07/07 and 03/08, terminated in 2012, COL application withdrawn 07/15	Refused offered lo guarante needs US equity
Green River, UT	AP1000	2 x 1250	Blue Castle / Transition Power Development		2030
River Bend, LA	ESBWR	1600	Entergy	25/09/08, COL application withdrawn	
South Texas Project[e], TX	ABWR	2 x 1356	Toshiba, NINA, STP Nuclear (merchant plant)	COLs issued Feb 2016 but design certification application withdrawn July 2018	Cancelle May 201
Nine Mile Point, NY	US EPR	1710	UniStar Nuclear (merchant plant)	30/09/08, COL application	

Site	Technology	MWe gross	Proponent/utility	Licensing status	Status
				withdrawn 2013	
Stewart County, GA	AP1000	1250	Georgia Power (Southern Co)	COL application deferred in 2017	Build aft 2030
William States Lee III, SC	AP1000	2 x 1250	Duke Energy	13/12/07, COL issued Dec 2016	Plans cancelle Aug 201
Idaho National Laboratory, ID	Aurora	1.5	Oklo Power	COL application lodged March 2020; denied Jan 2022	Oklo inte to submi new COL applicati for a different design o to 75 MV
Carbon Free Power Project, ID	NuScale US460	6 x 77	Utah Associated Municipal Power Systems (UAMPS)	COL application planned to be submitted Jan 2024	Project terminat Nov 202

Electricity market challenges

About 54 GWe of US nuclear capacity is in regulated markets, and 45 GWe in deregulated merchant markets, with power sold competitively on a short-term basis. In these liberalized markets, regional transmission organizations (RTOs) and independent system operators (ISOs) operate the grid, using free-market auctions and longer-term power purchase agreements under federal arrangements and rules. See Nuclear Energy Institute's list of nuclear plants in regulated and deregulated states.

In states with deregulated electricity markets, nuclear power plant operators have found increasing difficulty with competition on two fronts: low-cost gas, particularly from shale gas developments; and subsidized wind power with priority grid access. The imposition of a price on carbon dioxide emissions would help in competition with gas and coal, but this is not expected in the short-term. Single-unit plants which tend to have higher operating costs per MWh are most vulnerable. The basic problem is low natural gas prices allowing gas-fired plants to undercut power prices. A second problem is the federal production tax credit of $23/MWh paid to wind generators, coupled with their priority access to the grid. When there is oversupply, wind output is taken preferentially.

Capacity payments can offset losses to some extent, but where market prices are around $35-$40/MWh, nuclear plants are struggling. According to Exelon, the main operator of merchant plants and a strong supporter of competitive wholesale electricity markets, low prices due to gas competition are survivable, but the subsidized wind is not. Although wind is a very small part of the supply, and is limited or unavailable most of the time, it has a major effect on electricity prices and the viability of base-load generators.

A significant ISO for nuclear plants is PJM Interconnection which serves all or parts of 13 mid-Atlantic states and DC. In May 2014 five Exelon reactors at three plants – Oyster Creek, Quad Cities and Byron – for the first time failed to clear the PJM capacity auction for three years ahead, 2017-2018, so did not receive capacity payments or an assured market for 12 months, despite having been a reliable basis of supply in New Jersey and Illinois for decades, and of zero-carbon sources. Following the 2014 auction, FERC said it was actively considering ways it can ensure that base-load power sources, such as nuclear plants, are appropriately valued and their viability maintained in wholesale electricity markets. FERC's focus is on capacity markets and how they should take into account the full value of a base-load power plant; and on whether there are appropriate incentives for plants that contribute to the country's electric reliability in order for them to survive and continue providing those services.

In May 2017 Exelon's Three Mile Island (TMI) unit 1 and Quad Cities 1&2 failed to clear the PJM Interconnection capacity auction for 2020-21. Its other plants did clear in the auction, which cleared about $25 per megawatt-day below the previous year and $15 below market expectations at $76.53/MWd for the majority of the PJM footprint due to lower load forecasts and other factors. Exelon said that its nuclear units cleared a total of 13,275 MWe of capacity in the auction. Clearing prices for that capacity ranged from $188/MWd in the ComEd region serving Chicago, where Quad Cities is located, to $77/MWd in the RTO region. In TMI's region, the price was $88/MWd. Exelon said that TMI 1 had not cleared the past three PJM auctions and had not been profitable in five years. While the continued operation of Quad Cities was ensured by newly-introduced legislation in Illinois, Exelon warned that the TMI reactor, which entered service in 1974, was at risk of early retirement.

In May 2018, PJM's 2021-22 capacity market auction cleared at $140/MWd, an 83% increase over the 2017 auction. Despite the higher price, just 19 GWe of nuclear cleared, a decrease of 7.4 GWe from the previous year. Exelon said that TMI 1, Dresden and "all but

a small portion" of its Byron plant failed to clear. FirstEnergy, despite announcing retirement plans for 4 GWe of nuclear capacity in March, was required to offer the units into the auction – but none cleared. Exelon shut down TMI 1 in September 2019, but in July 2024 Constellation was in talks to restart the unit.

In May 2021, PJM's 2022-23 capacity market auction cleared at $50/MWd, well down on the 2021-22 auction due to a lower load forecast among other factors. Despite the lower price, nuclear utilities cleared an additional 4.5 GWe compared to the previous auction.

Early reactor retirements

In November 2015 Exelon said that its Clinton, Ginna and Quad Cities plants were at greatest risk of early retirement for economic reasons, with a question mark also over Byron. In May 2016 Exelon said it would close Clinton in June 2017 and Quad Cities in June 2018 unless the state of Illinois made provision for them to be profitable, by means of zero emission credits, likely to be capped at 20 TWh/yr for the 2884 MWe. New York state is making similar provision for its upstate plants (see below).

In June 2016 Omaha Public Power decided to close Fort Calhoun in Nebraska, the smallest US nuclear power plant, at the end of the year. PG&E in June 2016 announced that the Diablo Canyon units would close in 2024 and 2025. In March 2023 the NRC approved PG&E's request to operate the two units at Diablo Canyon past their respective 2024 and 2025 licence expiry dates on the condition that PG&E submitted licence renewal applications for the units by the end of 2023. The application for the two units was accepted by the NRC in December 2023.

Early in 2017 Entergy and the state of New York agreed that unit 2 of the Indian Point plant would close by the end of April 2020, followed by unit 3 in April 2021. Entergy cited "sustained low current and projected wholesale energy prices that have reduced revenues, as well as increased operating costs" coupled with political pressure. Entergy had invested over $1.3 billion in the two reactors over the 15 years it owned them. Its application for licence renewal of the two units was proceeding very slowly through the NRC review. In September 2018 the NRC approved Entergy's request to shorten the term of renewed operating licences for units 2 and 3 to 2024 and 2025 respectively. Unit 2 closed on 30 April 2020, and unit 3 on the same day a year later.

In September 2018 Exelon's single-reactor Oyster Creek plant in New Jersey was shut down, 11 years before its operation was due to end, so as to avoid the expense of state environmental regulations that would require the construction of $800 million cooling towers.

In May 2019, Entergy's 677 MWe single-reactor Pilgrim plant in Massachusetts was shut down due to market conditions and increased costs, the same situation as caused Entergy to close its 635 MWe Vermont Yankee reactor at the end of 2014, and plan to close its 852 MWe Fitzpatrick reactor in January 2017.

In August 2020, Exelon announced that it intends to retire its Byron and Dresden plants in Autumn 2021. Units 2&3 of the Dresden plant are licensed to run for a further 10 years, and units 1&2 of the Byron plant are licensed to run for a further 20 years. Exelon stated that the plants face revenue shortfalls amounting to "hundreds of million dollars" due to declining energy prices and market rules that allow fossil fuel plants to underbid clean resources in the PJM capacity market. Exelon also stated that its LaSalle and Braidwood plants were also at risk of premature closure. However, in September 2021 a new energy bill was signed into law in Illinois, which introduced $694 million in nuclear subsidies to be paid over 5 years*. Exelon subsequently announced that it was to refuel its Byron and Dresden plants.

* The bill also included subsides of more than $350 million annually for renewables.

Prematurely retired reactors

Reactors	State	Net capacity (MWe)	Shutdown
Crystal River 3**	FL	860	2013
San Onofre 2&3**	CA	1070, 1080	2013
Kewaunee	WI	566	2013
Vermont Yankee	VT	605	2014
Fort Calhoun	NE	482	2016
Oyster Creek 1	NJ	619	2018
Pilgrim 1	MA	677	2019
Three Mile Island 1	PA	819	2019
Indian Point 2	NY	998	2020
Duane Arnold	IA	601	2020

Reactors	State	Net capacity (MWe)	Shutdown
Indian Point 3	NY	1030	2021
Palisades*	MI	805	2022
Total		11,092	

Source: Nuclear Energy Institute
* Palisades was sold to Holtec for decommissioning. Holtec plans to restart the unit and has secured a $1.52 billion loan from the US DOE Loan Programs Office.
** Crystal River 3 closed after the operator, Duke Energy, decided against trying to repair a delamination within the containment concrete that had been discovered during uprate work. San Onofre 2&3 closed due to faults with the steam generators that were installed a year prior as part of an uprate programme at the plant.

Plants saved from premature retirement

Reactors	State	Net capacity (MWe)	Initially announced shutdown year
Beaver Valley 1&2	PA	908, 905	2021
Byron 1&2	IL	1164, 1136	2021
Clinton	IL	1062	2017
Davis-Besse	OH	894	2020
Dresden 2&3	IL	894, 879	2021
FitzPatrick	NY	813	2017
Hope Creek & Salem 1&2	NJ	1172, 1169, 1158	2020-2021
Millstone 2&3	CT	869, 1210	2020
Nine Mile Point 1&2	NY	613, 1277	2017-2018
Perry	OH	1240	2020
Quad Cities 1&2	IL	908, 911	2018
R. E. Ginna	NY	560	2017
Total		19,742	

Source: Nuclear Energy Institute

EPA Clean Power Plan

In June 2014 the US Environmental Protection Agency (EPA) announced that it would use its authority under the Clean Air Act to require a reduction in carbon emissions from US power plants of 25% below 2005 levels by 2020, and more by 2030, with states to be responsible for achieving this. There had already been a 16% drop since 2005. In August 2015 the EPA issued its Clean Power Plan to curb greenhouse gas emissions from existing fossil fuel-fired power plants under section 111(d) of the Clean Air Act and to reduce CO_2 emissions by 32% from 2005 levels by 2030. The Plan became effective in December 2015, and states were to have until September 2018 to submit their plans to comply with the emission reductions, using various means including increased thermal efficiency by 2.1-4.3%, greater use of nuclear power and renewables, and greater use of gas.

In November 2014 the National Association of Regulatory Utility Commissioners urged the EPA, in its proposed Clean Power Plan, to adopt regulations which "encourage states to preserve, life-extend, and expand existing nuclear generation." In January 2015 the Nuclear Energy Institute said that a top priority was for nuclear plant operators to be fully compensated in competitive wholesale US electricity markets for the value they provide as the main source of reliable, carbon-free, continuous base-load power. However, the majority of these measures were not included and the Clean Power Plan was heavily biased to wind and solar renewables. It allowed credit for new nuclear power plants and uprates to existing units, but would not credit the role of existing nuclear capacity, some of which is marginal economically in present market conditions. Nor would it credit nuclear licence extensions on the same basis as new capacity. Nuclear power produces about 55% of US carbon-free electricity, nuclear plants are already the main carbon-free generation source for over half of US states, and they avoid the emission of over 750 million tonnes of CO_2 per year relative to coal. It is accepted that the 32% CO_2 reduction by 2030 will be impossible without at least the present level of nuclear contribution. About one-third of the nation's 300 GWe of coal-fired base-load capacity is expected to be retired by 2030. Some states were preparing legal challenges to the Plan, others remain committed to it.

In March 2017 President Trump signed the Energy Independence Policy executive order which aimed to roll back the 2015 EPA Clean Power Plan, and called for the EPA to review it to remove what may

“unduly burden the development of domestic energy resources.” The impact of this could not be immediate, and may be more in tone than substance. It would take several years under notice and comment rulemaking processes, and the main timeline under the Plan was 2030 in any case. US electricity should be "affordable, reliable, safe, secure, and clean,” presumably in that order of priority. The executive order rescinded several climate change measures. In October 2017 the EPA issued a notice of proposed rulemaking (NOPR) to repeal the Clean Power Plan on the grounds that it exceeds the EPA’s authority under the Clean Air Act and sets emissions standards that power plants could not reasonably meet. Repeal of the plan, which was premised on a “novel and expansive view of Agency authority,” would save $33 billion in compliance costs by 2030 according to the EPA. The plan was repealed in June 2019.

In November 2020 the USA formally withdrew from the Paris Agreement. On 20 January 2021, the first day of the Biden administration, the country rejoined the agreement. On 20 January 2025, President Trump once again withdrew the United States from the agreement.

Regional Greenhouse Gas Initiative

The Regional Greenhouse Gas Initiative is a 2009 cap-and-trade programme for reducing carbon dioxide emissions, covering fossil-fuel plants above 25 MWe in the northeast and mid-Atlantic states of Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Rhode Island and Vermont. Virginia participated from 2021 to 2023 but withdrew. Pennsylvania’s regulations to join were repealed in November 2025. Carbon dioxide emissions allowances are auctioned quarterly, with prices around $27/tonne as of late 2025.

Electricity market reforms

State initiatives, zero-emission credits

A number of states are taking action to counteract problems with the markets, which the states do not control, to preserve values not recognized in the markets.

New York

In December 2015 the New York state governor directed its Department of Public Service (NYDPS) to develop a clean energy standard (CES) that calls for a 40% reduction in greenhouse gas emissions from 1990 levels by 2030 and a longer-term decrease of 80% by 2050, while not losing carbon reduction gains achieved to date. The state intended to comply with the EPA Clean Power Plan, and its six nuclear reactors provided nearly one-third of the state's electricity in 2015. Entergy had announced the premature closure of its FitzPatrick nuclear plant in upstate New York by January 2017, and Exelon had warned its Ginna and Nine Mile Point plants were at risk of closure for similar economic reasons. The governor said that closing nuclear facilities "would eviscerate the emission reductions achieved through the state's renewable energy programmes, diminish fuel diversity, increase price volatility, and financially harm host communities." The New York independent system operator later warned that to preserve the reliability of the grid, the state must keep all of its nuclear plants operating while slowing renewable energy growth.

The NYDPS issued a white paper in January 2016 proposing 'zero-emission credits' (ZECs) for nuclear generators that would work in parallel with the tax credits that renewable sources receive, and provide the market signals necessary to warrant continued operation of these non-emitting plants. The Nuclear Energy Institute noted that the proposal "establishes a mechanism that can ensure nuclear operators receive the market signals necessary to warrant continued operation of these non-emitting assets." In addition, a cost study issued by the NYDPS in April 2016 as a supplement to the white paper showed the "outstanding value" that including nuclear in the clean energy standard would provide to New York citizens. The study pointed out that the zero-emission credits would generate $2.8 billion in benefits, or two-thirds of the entire clean energy standard programme's $4.4 billion, for $270 million (less than 8% of the programme's costs).

In July 2016 the NYDPS put forward a proposal which would value the zero-emissions attributes of the upstate nuclear power plants (*i.e.* not including Indian Point), based on the social cost of carbon and requiring the distribution utilities "to pay for the intrinsic value of carbon-free emissions from nuclear power plants by purchasing zero-emission credits." The department said that there is a "public necessity" for subsidies for the Fitzpatrick, Ginna and Nine Mile Point plants. The benefits of paying such subsidies would far

outweigh the costs, the department said. During the first two years of the programme, the state's economic and environmental benefits associated with carbon reductions, supply cost savings and property tax benefits were estimated to be about $5 billion, against total payments of up to $965 million – a net benefit of $4 billion.

The NY Public Service Commission on 1 August 2016 approved the CES plan but excluded Indian Point. The majority vote was reported to be on three main criteria: grid reliability, reducing carbon emissions, and maintaining jobs. The governor's announcement said: "A growing number of climate scientists have warned that if these nuclear plants were to abruptly close, carbon emissions in New York will increase by more than 31 million metric tons during the next two years, resulting in public health and other societal costs of at least $1.4 billion."

New York's ZEC programme is being implemented in six tranches over a period of 12 years from April 2017. For the first two-year period nuclear generators received ZECs of $17.54/MWh, paid by the distribution utilities (and hence eventually ratepayers) but otherwise similar to the federal production tax credits (PTC) applying to renewables since 1993 on an inflation-adjusted basis, though at a lower rate than its $23/MWh for wind. ZECs will escalate to $29.15/MWh over subsequent years. Later, in July, Entergy's Indian Point plant was included in the proposal, albeit not for the first two years.

The broader CES required that NY state's utilities source at least half their electricity from renewables by 2030, less than it gets now from all clean energy sources: nuclear 32%, hydro 19%, wind 3%, and solar (less than 1%). Gas supplies 40% of power. In 2019 the CES was revised to require 100% carbon-free electricity by 2040.

In August 2016 Exelon agreed to buy the 838 MWe Fitzpatrick plant, which is licensed to 2034, from Entergy for $110 million in anticipation of the NYPDS CES proposal being implemented. It also confirmed that it would proceed with investing about $200 million in Nine Mile Point and Ginna plants early in 2017 and would "invest hundreds of millions of dollars in Fitzpatrick in January to refuel the plant and upgrade systems needed to reverse the shutdown decision." Entergy said it plans "to move away from merchant power markets and toward a company operating exclusively as a utility in regulated markets."

In October 2016 a coalition of non-nuclear energy companies and groups filed a lawsuit against the New York Public Service Commission challenging the PSC's authority to raise electricity rates

to pay for the ZECs which will subsidize the continued operation of several nuclear power plants. The plaintiffs, led by the Coalition for Competitive Electricity, included Dynegy, Eastern Generation, Electric Power Supply Association, NRD Energy, Roseton Generating and Selkirk Cogen Partners. This legal challenge failed, and an appeal to the Supreme Court challenging the ZEC programme was rejected in April 2019.

In January 2026 ZECs for Constellation's Fitzpatrick, Ginna and Nine Mile Point plants were extended from 2029 to 2049. This action was taken early to give Constellation confidence to invest in licence renewals for Nine Mile Point 1 and Ginna, which it planned to submit in the first quarter of 2026.

Illinois

In February 2015 Illinois, another state with a deregulated market, took steps to enhance the competitiveness of nuclear power and renewables. The Illinois Low Carbon Portfolio Standard would require utilities to purchase low-carbon energy credits equivalent to 70% of their retail sales to customers within the state. This was congruent with the subsequent EPA Clean Power Plan. Eleven Exelon nuclear reactors at six sites supply almost half of the state's electricity. In mid-2016 the legislation had lapsed. Following the failure of Illinois legislature to pass its Next Generation Energy Plan, in June 2016 Exelon said that it would move forward with plans to close down Clinton in June 2017 and Quad Cities a year later. It would terminate capital investment projects required for the long-term operation of both plants, and would immediately take one-time charges of $150 million to $200 million for 2016, and accelerate some $2 billion in depreciation and amortization.

In October 2016 Exelon confirmed that it would close the Quad Cities and Clinton plants if legislation was not passed by year end since they had lost more than $800 million in the past seven years. In November the Future Energy Jobs Bill was introduced, reflecting "a diverse set of interests, as well as agreement in important areas among environmentalists, consumer advocates, community leaders and energy companies." A core feature of the legislation is the establishment of the Zero Emission Standard to preserve the state's two at-risk nuclear plants, saving 4200 jobs, retaining $1.2 billion economic activity annually and avoiding increases in energy costs. The bill provides ZECs similar to those in New York – "a tradable credit that represents the environmental attributes of one megawatt hour of energy produced from a zero

emission facility" such as the nuclear power plants which supply about 90% of the state's zero-carbon electricity. The state legislature passed the bill in December 2016. It will provide up to $235 million annually to support the two plants for ten years. The state utilities will purchase ZECs from the nuclear generators and collect payments from ratepayers. The legislation sets the value of a ZEC to be $16.50/MWh based on the social cost of carbon.

A legal challenge to the Illinois ZEC programme failed, and in January 2019 a coalition of power generation companies took the appeal to the Supreme Court, where it was rejected.

In August 2019 Exelon said that its Braidwood, Byron and Dresden nuclear plants in the state were "financially challenged" and that the company was working with state lawmakers to ensure that they were included in any legislation that supports clean energy sources. In August 2020 Exelon said it planned to permanently close the Byron and Dresden nuclear power plants in September 2021 and November 2021, respectively. However, the premature retirement of the two plants was averted following the introduction of a new energy bill in September 2021 (see above).

Ohio

In February 2017 FirstEnergy announced that it was in dialogue with the Ohio state government to try to secure the future of its two nuclear plants in the state, Davis-Besse and Perry, a 894 MWe PWR and a 1256 MWe BWR respectively, owned by its subsidiary FirstEnergy Solutions *(Beaver Valley just over the border in Pennsylvania is excluded)*. The company had earlier announced its intention to withdraw from competitive generation markets by mid-2018, and in the fourth quarter of 2016 recorded a $9.2 billion impairment charge as a result.

In October 2017 a new bill was introduced into Ohio legislature aiming to establish the Zero Emissions Nuclear (ZEN) programme to support the state's two nuclear plants. The bill stated an initial ZEC price of $17/MWh. Each participating utility would be limited to purchasing one-third of its recorded 'total end user consumption' in MWh over the previous two calendar years.

FirstEnergy had 13,000 MWe of generating capacity operating in deregulated markets. It decided to relinquish all these assets by mid-2018, and withdraw from competitive generation altogether, maintaining only its generation assets in regulated markets. Due to competition from low-cost gas and subsidized wind power, the

units were unlikely to be sellable if states failed to introduce legislation to provide zero emission credits. In March 2018, with the proposed Ohio bill stalled in a Senate committee, FirstEnergy filed a deactivation notice for its David-Besse and Perry plants, as well as its Beaver Valley plant in Pennsylvania. The deactivation notice set retirement dates of 2020 for Davis-Besse, and 2021 for Perry and Beaver Valley. FirstEnergy stated that it would continue to work with officials from the two states, and called on them to consider policy solutions to prevent early closure of the assets.

In May 2019 a bill (Ohio House Bill 6, HB6) creating the Clean Air Program passed Ohio's lower house. HB6 was approved by Ohio legislature and signed into law on 23 July 2019. It establishes credits for certified clean air resources, including nuclear plants, at $9/MWh. Under the bill, Ohio's electric distribution utilities collect a monthly charge capped at $0.85 from retail electric customers, and up to $2400 for large industrial plants, to fund payments to generators. Following the passing of the bill, FirstEnergy halted the deactivation orders for Davis-Besse and Perry. Several bills to repeal HB6 have since been introduced following the arrest in July 2020 of the Speaker of Ohio's House of Representatives and several others on charges of bribery to pass the legislation.

FirstEnergy Solutions filed for bankruptcy in March 2018 and in February 2020 it separated from its parent company when it emerged from bankruptcy protection as Energy Harbor.

Connecticut

In March 2017 Connecticut's Energy & Technology Committee approved a bill supporting the continued operation of Dominion's Millstone plant in that deregulated market. The bill "would expand the state's existing renewable electricity procurements to nuclear power by directing state regulators to solicit up to half of the facility's annual generation (*i.e.* 8.3 TWh) for five-year power purchase agreements." In October 2017, Connecticut's legislature passed the bill, supporting the continued operation of Millstone. After a 23:8 Senate vote, the lower house passed the bill 75:66. It made Dominion eligible to bid for long-term supply contracts for up to half of Millstone's output as a clean-energy resource, at higher prices, subject to the state Department of Energy and Environmental Protection and Public Utilities Regulatory Authority determining that this is in the public interest. The plant is the largest in New England and its viability has been eroded by cheap natural gas. Closure of the plant, which provides half of the state's power

and almost all of its zero-carbon power, would jeopardize the state's ability to meet its long-term goals for reducing carbon emissions. In December 2018, the Public Utilities Regulatory Authority agreed that the Millstone nuclear plant was at risk, allowing it to take part in zero-emission energy auctions. In March 2019 the plant obtained a 10-year contract for 9 TWh per year with two utilities. The two units operating at Millstone – units 2&3 – are licensed to 2035 and 2045.

Kentucky

In March 2017 Kentucky voted to end its moratorium on nuclear power in the state.

Pennsylvania

In March 2017 Pennsylvania set up a bipartisan, bicameral nuclear energy caucus to secure the role of nuclear energy in the state, where it provides about 40% of the electricity and contributes $2.3 billion to the state GDP. There are several two-unit nuclear power plants in the state: Beaver Valley, Limerick, Peach Bottom and Susquehanna. Three Mile Island shut down in September 2019. Prior to its shutdown, Exelon said that the 890 MW Three Mile Island 1 was "economically challenged as a result of continued low wholesale power prices and the lack of federal or Pennsylvania energy policies that value zero-emissions nuclear energy."

A draft law updating the Pennsylvania Alternative Energy Portfolio Standards Act to include nuclear energy was introduced to the state's legislature in March 2019. Despite nuclear power's importance to the state, it is excluded from the AEPS programme. The Keep Powering Pennsylvania Act would offer subsidies to nuclear plants and was put forward as costing $500 million per year, significantly less than the cost if economically-challenged plants were to close. Plants applying to join the programme need to agree to operate for at least six years. The bill had not been passed by the time Exelon needed to decide on Three Mile Island's future.

New Jersey

In April 2018, New Jersey legislators passed bills establishing a ZEC programme. In April 2019 the New Jersey Board of Public Utilities

(NJBPU) awarded ZECs to the Salem and Hope Creek nuclear power plants. The programme is to be funded by a 0.4 c/kWh tariff imposed on retail distribution customers. The bill requires plants to be licensed to operate until at least 2030, so excluded Exelon's Oyster Creek. Public Service Enterprise Group (PSEG), which operates the Hope Creek and Salem plants, had previously warned that closures were likely without intervention. The government expects that the two plants would receive about $200 million per year in revenue from ZEC sales to public utilities, apparently at around $10-11/MWh. The Oyster Creek plant (619 MWe net) closed in September 2018. Hope Creek 1 and Salem 1&2 are eligible to receive ZECs between April 2019 and May 2022. In April 2021 the NJBPU awarded a three-year extension, to 2025, for both plants. The ZEC programme ended in June 2025, with Inflation Reduction Act tax credits serving as replacement support.

In June 2017 MIT's Center for Energy and Environmental Policy Research published a new study that found that saving US nuclear "would come at a cost of $4-7/MWh on average in these markets, which is much lower than the cost of subsidizing wind power." The current production tax credit (PTC) level for renewables is $23/MWh.

Department of Energy rulemaking

Using its legislated authority for the first time since 1979, in September 2017 the Department of Energy (DOE) directed the Federal Energy Regulatory Commission (FERC) through a notice of proposed rulemaking (NOPR) to ensure that the country's "diverse mix of resources must include traditional base-load generation with onsite fuel storage that can withstand major fuel supply disruptions caused by natural and man-made disasters." The DOE said that FERC had so far "not done enough to address the crisis at hand" caused by the premature retirement of reliable plants. "Immediate action is necessary to ensure fair compensation in order to stop the imminent loss of generators with onsite fuel supplies, and thereby preserve the benefits of generation diversity and avoid the severe consequences that additional shutdowns would have on the electric grid," the DOE said in the NOPR. In particular, "the continued loss of base-load generation with onsite fuel supplies, such as coal and nuclear, must be stopped."

In January 2018 FERC halted the NOPR and called on operators of regional wholesale markets to "provide information as to whether the FERC and the markets need to take additional action on

resilience of the bulk power system." This removed the built-in incentives for coal and nuclear plants outlined in the September NOPR which would have required independent system operators and regional transmission organizations "to ensure that certain reliability and resiliency attributes of electric generation resources are fully valued." In particular, it stated that eligible "fuel-secure generation units", which are frequently relied upon for grid reliability and resilience, must be able to fully recover their costs.

Transmission infrastructure

The USA has a patchwork of grids which are often barely interconnected. The Western Interconnection includes about 11 states plus British Columbia and Alberta. ERCOT (Electric Reliability Council of Texas) includes most of Texas, and Eastern Interconnection takes in the rest of the USA and Canada. There is very little grid capacity in the middle of the country. Exelon has temporarily curtailed off-peak output at one or more of its nuclear plants in Illinois numerous times for more than a year to late 2016 because of grid constraints. The company has previously said intermittent grid congestion has been occurring in the region around those plants because of transmission line outages for scheduled maintenance, large influxes of wind-generated power into the grid during off-peak hours, or a combination of those factors.

There is an evident need for major investment, and in August 2017 the DOE Staff Report to the Secretary on Electricity Markets and Reliability recommended that the Federal Energy Regulatory Commission (FERC) takes a leading role in ensuring effective grid connections to meet base-load demand more widely and reliably. See above section on Department of Energy rulemaking.

More information on the US grid situation is in the information paper on Electricity Transmission Grids.

Consolidation of ownership and management

The US nuclear power industry underwent significant consolidation in the early 2000s, driven largely by economies of scale, deregulation of electricity prices and the increasing attractiveness of nuclear power relative to fossil generation. As of the end of 1991, a total of 101 individual utilities had some (including minority)

ownership interest in operable nuclear power plants. At the end of 1999, that number had dropped to 87, and the largest 12 of them owned 54% of the capacity. With deregulation of some states' electricity markets came a wave of mergers and acquisitions in 2000-1 and today the top 10 utilities account for more than 70% of total nuclear capacity. The consolidation has come about through mergers of utility companies as well as purchases of reactors by companies wishing to grow their nuclear capacity.

In respect to the number of operators of nuclear plants, this dropped from 45 in 1995 to about 30 in 2025, showing a substantial consolidation of expertise.

Mergers and consolidation of management

Most of the of nuclear generation capacity involved in consolidation announcements has been associated with corporate mergers, some of which failed due to regulatory opposition. Another means of consolidation has been via management contracts, and other means of management rationalization for single-unit plants have also occurred. Details are in Appendix 2: Power Plant Purchases.

Purchase of reactors

In the 12 years from 1998, there were 20 reactor purchase deals involving 25 plants, usually in states where electricity pricing had been deregulated (see Nuclear Power in the USA Appendix 2: Power Plant Purchases). The plants acquired were often those with high production costs, offering the potential for increased margins if costs could be reduced. Of the 5900 MWe involved to mid-2000, half was associated with plants having 1998 production costs above 2.0 cents per kWh. Sellers tended to consider the higher-cost plants as potential liabilities and were willing to get rid of them for a fraction of their book value, whereas the larger utility buyers considered the plants to be potential assets, depending only on their ability to lower the production costs. In many cases, large power companies acquired plants from local utility companies and at the same time entered contracts to sell electricity back to the former owners. Entergy Corporation, for example, bought two reactors from New York Power Authority in 2000 and agreed to make the first 500 MWe of combined output available at 2.9 cents/kWh and the remainder at 3.2 or 3.6 cents/kWh.

Along with Exelon, Entergy is a prominent example of the consolidation that occurred. Originally based in Arkansas, Louisiana, Mississippi and eastern Texas, Entergy doubled its nuclear generation capacity over 1999 to 2007 with the acquisition of reactors in New York, Massachussets, Vermont and Michigan, as well as a contract to operate a nuclear plant in Nebraska. Other companies that have increased their nuclear capacity through plant purchases are FPL Group based in Florida (four units), Constellation Energy based in Maryland (three units, since merged with Exelon) and Dominion Resources based in Virginia (four units).

However, some older plants acquired from their original owners for their value as 'cash cows' are now unprofitable in deregulated markets and threatened with closure due to very low natural gas prices. In addition, onerous safety requirements following the Fukushima accident compound the economic challenges with already tight NRC regulations. See comments above (in the section on State initiatives zero-emission credits) regarding some Exelon and Entergy plants in deregulated markets.

Notes & references

Notes

a. The first nuclear reactor in the world to produce electricity (albeit a trivial amount) was the small Experimental Breeder Reactor (EBR-1) in Idaho, which started up in December 1951. In 1953, President Eisenhower proposed his *Atoms for Peace* programme, which reoriented significant research effort towards electricity generation and set the course for civil nuclear energy development in the USA. The Mark 1 naval reactor of 1953 led to the US Atomic Energy Commission building the 60 MWe Shippingport demonstration PWR reactor in Pennsylvania, which started up in 1957 and operated until 1982. [Back]

b. Fort St. Vrain in Colorado was a 330 MWe high-temperature gas-cooled reactor (HTGR) operating 1976-89. The technology was developed from an earlier 40 MWe HTGR at Peach Bottom, Pennsylvania, which operated from 1967 to 1974. [Back]

c. The NRC had approved full design certification for the Westinghouse AP1000 in 2005 and issued a final rule certifying the design in January 2006. However, in May 2007, Westinghouse

submitted an application to amend the AP1000 final design certification rule. [Back]

d. The ABWR design that has NRC certification is the GE Hitachi design, some aspects of which are proprietary to GE Hitachi. While the licence application for the first new ABWRs to be announced for the USA – at the South Texas Project (STP) – references the certified GE Hitachi design, Toshiba was selected as the main contractor to build the units. In November 2010, Toshiba submitted an application to renew the design, which includes revisions to bring the certified design in line with the STP units (see Note j below). [Back]

e. An asterisk (*) denotes reference COL for reactor type. EPC = Engineering, procurement and construction agreement. Merchant plants are without regulated cost recovery. 'Planned' status shows a higher level of commitment – such as an order for large forgings or an EPC contract – than 'Proposed' status. [Back]

f. Dominion's North Anna COL application referenced the ESBWR, but in March 2009 it issued a new request for proposals from reactor vendors and in May 2010 it selected the Mitsubishi US-APWR. Then in April 2013 it reverted to the ESBWR, and agreed on an EPC contract for it with GE Hitachi and Fluor.

The COL reviews of Entergy's applications for Grand Gulf and River Bend, along with the review of Exelon's application for the Victoria County site were suspended by the NRC, following the decisions by Entergy and Exelon to review their initial reactor design choice of the ESBWR. Exelon had initially proposed two ESBWR units for its Victoria County site but, early in 2009, switched to the ABWR design, to be built by GE Hitachi. Shortly afterwards, citing adverse economic conditions, Exelon withdrew its COL application. [Back]

g. The site chosen by the NuStart Energy Development consortium for the reference COL application for the AP1000 was originally TVA's Bellefonte. However, NuStart later decided to transfer the AP1000 reference COL application to Vogtle on the grounds that the Vogtle application had "specific near-term construction plans." In May 2009, NuStart announced that it was "consulting with the Nuclear Regulatory Commission and Department of Energy to develop a process for transferring the reference combined construction and operating licence application from TVA's Bellefonte nuclear site to Southern Nuclear's Vogtle Electric Generating Plant."[1] [Back]

h. A COL application for two proposed AP1000 units as units 3&4 at TVA's Bellefonte site was submitted to the Nuclear Regulatory

Commission in October 2007. This COL application was originally the reference COL application for the AP1000 design but the reference application was transferred to Vogtle. The site also has two unfinished 1213 MWe PWRs (unit 1 being about 88% complete and unit 2 about 58% complete) and TVA has been considering all options for the site, including the completion of units 1&2. In May 2010 theTVA staff identified completion of unit 1 as the best option for the site, and in August 2011 the TVA Board decided to complete unit 1.[2] [Back]

i. AmerenUE announced in April 2009 that it was suspending its efforts to build a new unit and in June 2009 the company requested the Nuclear Regulatory Commission to suspend all review activities relating to the Callaway 2 COL application. However, in April 2012 Ameren Missouri set out to seek DOE support for the first of five Westinghouse SMR units at Callaway. In July 2015 Ameren withdrew its COL application. [Back]

j. Since the decision to go ahead with South Texas Project (STP) units 3&4 was first announced, there have been a number of developments. The combined construction and operating licence (COL) application was prepared by STP Nuclear Operating Company (STPNOC) together with GE Hitachi Nuclear Energy and Bechtel and submitted in September 2007.[3] Just before submittal of the COL application, NRG Energy and STPNOC signed a project services agreement with Toshiba to support the design, engineering, construction and procurement of the units. Fluor was then enrolled to support Toshiba[4]. In November 2010, Nuclear Innovation North America LLC (NINA, the nuclear development company jointly owned by NRG Energy and Toshiba) announced that it had awarded the engineering, procurement and construction (EPC) contract to a "restructured EPC consortium" of Toshiba's US subsidiary Toshiba America Nuclear Energy Corporation (TANE) and The Shaw Group[5] (later CB&I). Following CB&I's sale of its CB&I Stone & Webster subsidiary to Westinghouse (then owned by Toshiba), in May 2016 Toshiba and CB&I dissolved their 2010 partnership in relation to all ABWR plans, leaving TANE as the sole EPC contractor for the project.

In the meantime, the reactor technology moved from being based on the GE design certified by the US Nuclear Regulatory Commission in 1997. The design had to be renewed by 2012 and a renewal application by Toshiba was submitted in November 2010.[6] The renewal application included updates and revisions in accordance with the STP design. Hence, the STP reactors were considered to be Toshiba ABWRs, whereas the original intention was to use the 1997 certified design "with only a limited number of

changes to enhance safety and construction schedules," with these changes incorporated into the COL application[7]. However, in 2016 Toshiba's application for design certification renewal was withdrawn. [Back]

k. The COL review by the NRC was due to be completed late in 2011, and the units were expected online in 2016 and 2017, but in late 2011 the NRC notified NINA that the corporation did not meet the foreign ownership requirements and would therefore be ineligible to receive a licence; however NINA subsequently filed revisions to its COL application and a "negation action plan" to address the issue. In April 2013 the NRC "determined that NINA and its wholly owned subsidiaries ... continue to be under foreign ownership, control, or domination and do not meet the requirements ... of the Atomic Energy Act or the requirements of (federal regulations)." The NRC decision was reviewed by the NRC Atomic Safety Licensing Board (ASLB), which ruled in April 2014 that the 10% Toshiba equity was no problem. NRC's Advisory Committee on Reactor Safeguards in April 2015 also supported issuing the COLs and the NRC issued a final safety evaluation report in September 2015. In February 2016 the NRC issued the COLs. [Back]

l. To the end of September 2010, the Nuclear Regulatory Commission (NRC) had approved 135 power uprates totalling 5810 MWe (not including capacity recapture uprates for provisional operating licence plants) and this had increased to 7921 MWe (164 uprates) as of October 2020. Information on power uprates is available on the NRC website. [Back]

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Appendices

Nuclear Power in the USA Appendix 1: US Operating Nuclear Reactors
Nuclear Power in the USA Appendix 2 Power Plant Purchases
Nuclear Power in the USA Appendix 3: COL Applications

Related information

USA: Nuclear Fuel Cycle
USA: Nuclear Power Policy